

Sunflower-Free Uniform Families: Recursive Constructions and Explicit Bounds

Edward Axante*, Cristian Budala*, David Chitic*, Bogdan Dumitru* and Mihai Nacu*

Abstract

Let $f(w, k)$ be the maximum size of a w -uniform family containing no sunflower with k petals. We introduce a recursive construction for sunflower-free families and use it to obtain a general lower bound on the exponential growth rate of $f(w, k)$. We also prove a general upper bound for 3-uniform families with at least four petals. Our results give $39 \leq f(3, 4) \leq 49$, $f(3, 5) \leq 146$, $153 \leq f(3, 6) \leq 255$, $259 \leq f(3, 7) \leq 474$, and $54 \leq f(4, 3) \leq 83$. In addition, we prove that the maximum size of an intersecting 4-uniform family containing no sunflower with three petals is 27. The upper bounds 49 and 83 are computer-assisted. The finite lower bounds come from explicit constructions.

1 Introduction

A sunflower with k petals, or k -sunflower, is a collection of k distinct sets whose pairwise intersections are equal. Their common intersection is called the core and may be empty. Let $f(w, k)$ denote the maximum size of a family of distinct w -element sets containing no k -sunflower. The sunflower conjecture of Erdős and Rado asks whether, for every fixed k , there is a constant C_k such that $f(w, k) \leq C_k^w$ for every w [11]. Alweiss, Lovett, Wu, and Zhang substantially improved the classical upper bound [6], and Bell, Chueluecha, and Warnke subsequently proved $f(w, k) \leq (Ck \log w)^w$ for an absolute constant C and all $w, k \geq 2$ [7].

Our lower-bound arguments keep track not only of the absence of sunflowers but also of the maximum number of pairwise disjoint members. Controlling this matching number rules out sunflowers with empty core. Within this framework, we construct larger sunflower-free families from smaller ones by attaching subsets of an auxiliary set. The resulting recurrence leads to an explicit general lower bound on the exponential growth rate of $f(w, k)$. An alternative construction based on a bounded-degree auxiliary graph is included in the appendix.

Abbott and Hanson introduced the study of sunflower-free families with a bounded matching number and showed how families on disjoint ground sets can be combined to obtain lower bounds for $f(w, k)$ [4]. For three petals, our main recurrence follows from a multiplication theorem of Abbott and Exoo [5]. We extend this recurrence to arbitrary numbers of petals and all admissible matching-number bounds. To our knowledge, this recurrence, the alternative recurrence in the appendix, and the resulting exponential lower bound are new. Abbott and Hanson raised the problem of proving recurrences for ψ and reported no success [4, p. 11]. The closest prior forms are an additive recurrence of Abbott and Exoo for three petals [5, Theorem 3] and an edge-attachment product of Frankl and Wang [13, Proposition 2.1]. For $k - 1 \geq 3w(m - 1)$, Frankl and Wang determine $\psi(w, k, m)$ exactly [13, Theorem 1.11]; the bounds proved here concern fixed k and growing w , outside that regime.

*Faculty of Mathematics and Computer Science, University of Bucharest, Romania.

For upper bounds on families of triples, we begin with a largest collection of pairwise disjoint triples and classify all other members by how they intersect its union. This reduces the problem to degree and matching bounds for ordinary graphs, to which we apply the theorem of Chvátal and Hanson [9]. The resulting argument gives a general upper bound for $f(3, k)$ when $k \geq 4$. Separate finite analyses sharpen the result for $f(3, 4)$ and give the upper bound for $f(4, 3)$.

Table 1 compares the numerical bounds obtained in this paper with previously published bounds. The two published endpoints shown for $f(3, 7)$ follow from Proposition 2.1 and Theorem 1.14, respectively, of Frankl and Wang [13].

Table 1: Published comparisons and bounds proved in this paper.

Quantity	Published comparison	Bound proved here
$f(3, 4)$	$38 [5] \leq f(3, 4) \leq 69 [3]$	$39 \leq f(3, 4) \leq 49$
$f(3, 5)$	$f(3, 5) \leq 180 [3]$	$f(3, 5) \leq 146$
$f(3, 6)$	$146 [5] \leq f(3, 6) \leq 305 [3]$	$153 \leq f(3, 6) \leq 255$
$f(3, 7)$	$252 \leq f(3, 7) \leq 498 [13]$	$259 \leq f(3, 7) \leq 474$
$f(4, 3)$	$54 [4] \leq f(4, 3) \leq 142 [3]$	$54 \leq f(4, 3) \leq 83$

The upper bounds $f(3, 4) \leq 49$ and $f(4, 3) \leq 83$ are computer-assisted and follow from exhaustive finite enumerations. The lower bounds $f(3, 4) \geq 39$ and $f(3, 7) \geq 259$ are established by explicit finite families. We also determine exactly the maximum size of an intersecting 4-uniform family containing no 3-sunflower: it is 27. The main combinatorial arguments of sections 2 to 4, the appendix capacity recurrence, and every explicit finite lower-bound witness have additionally been formalised and machine-checked in the Lean 4 proof assistant (section 8).¹

2 Notation and preliminary facts

This section fixes the notation and records the basic facts used in the paper. We introduce links and the matching-constrained extremal quantity, determine the cases of uniformity one and two, and recall the product inequality used to define the exponential growth rate.

2.1 Set families

All set families in this paper are finite and have distinct members. All graphs are finite and simple. For a finite set X , write

$$\binom{X}{w} = \{A \subseteq X : |A| = w\}.$$

A family $\mathcal{F}$ is w -uniform if $\mathcal{F} \subseteq \binom{X}{w}$ for some finite set X . Its support is

$$\text{supp}(\mathcal{F}) = \bigcup_{F \in \mathcal{F}} F.$$

¹Code, data, and the Lean development are available at <https://github.com/bogdan27182/sunflower-paper>.

If distinct sets $A_1, \dots, A_k$ satisfy

$$A_i \cap A_j = C \quad (1 \leq i < j \leq k),$$

then they form a k -sunflower with core C . The core may be empty. The sets $A_i \setminus C$ are its petals. For $w \geq 1$ and $k \geq 2$, define

$$f(w, k) = \max\{|\mathcal{F}| : \mathcal{F} \text{ is } w\text{-uniform and contains no } k\text{-sunflower}\}. \quad (2.1)$$

The Erdős–Rado sunflower lemma makes this maximum finite [11].

Lemma 2.1 (Occurrence rule). *Let $A_1, \dots, A_k$ form a sunflower. Every point in $A_1 \cup \dots \cup A_k$ belongs to exactly one member or to all k members.*

Proof. Suppose that a point belongs to two members. It then belongs to their intersection, which is the core. Hence it belongs to every member. $\square$

The core of a sunflower of distinct w -element sets has size at most $w - 1$. In particular, the possible core sizes for a sunflower of triples are zero, one, and two.

2.2 Matchings, degrees, and links

A matching in a family is a subfamily of pairwise disjoint members. The matching number $\nu(\mathcal{F})$ is the largest size of a matching in $\mathcal{F}$. A family is intersecting if every two distinct members meet. Equivalently, $\nu(\mathcal{F}) \leq 1$. Since a matching of size k is a k -sunflower with empty core, every k -sunflower-free family satisfies

$$\nu(\mathcal{F}) \leq k - 1. \quad (2.2)$$

For $m \geq 2$, define

$$\psi(w, k, m) = \max\{|\mathcal{F}| : \mathcal{F} \text{ is } w\text{-uniform, contains no } k\text{-sunflower, and } \nu(\mathcal{F}) \leq m - 1\}. \quad (2.3)$$

This matching-constrained quantity was introduced by Abbott and Hanson [4]. In particular, $\psi(w, k, 2)$ is the maximum over intersecting w -uniform families without a k -sunflower, while $\psi(w, k, k) = f(w, k)$.

For a set S , define its degree in $\mathcal{F}$ by

$$d_{\mathcal{F}}(S) = |\{F \in \mathcal{F} : S \subseteq F\}|.$$

For $S = \{x\}$, write $d_{\mathcal{F}}(x)$. The link of S is

$$\text{Lk}_{\mathcal{F}}(S) = \{F \setminus S : F \in \mathcal{F}, S \subseteq F\}.$$

Lemma 2.2 (Link bound). *Let $\mathcal{F}$ be a w -uniform family without a k -sunflower. If $|S| = s < w$, then $\text{Lk}_{\mathcal{F}}(S)$ is a $(w - s)$ -uniform family without a k -sunflower. Consequently,*

$$d_{\mathcal{F}}(S) = |\text{Lk}_{\mathcal{F}}(S)| \leq f(w - s, k). \quad (2.4)$$

Proof. The members of $\text{Lk}_{\mathcal{F}}(S)$ are distinct. If k of them formed a sunflower with core C , adjoining S would give a sunflower in $\mathcal{F}$ with core $C \cup S$. $\square$

Two families $\mathcal{A}$ and $\mathcal{B}$ are cross-intersecting if $A \cap B \neq \emptyset$ for every $A \in \mathcal{A}$ and $B \in \mathcal{B}$. A collection of families is pairwise cross-intersecting if each pair in the collection is cross-intersecting.

2.3 The graph case

For integers $D, q \geq 1$, let $\text{CH}(D, q)$ be the maximum number of edges in a graph with maximum degree at most D and matching number at most q . The Chvátal–Hanson theorem [9] gives

$$\text{CH}(D, q) = Dq + \left\lfloor \frac{D}{2} \right\rfloor \left\lfloor \frac{q}{\lceil D/2 \rceil} \right\rfloor. \quad (2.5)$$

Lemma 2.3 (Uniformities one and two). *For every $k \geq 2$,*

$$f(1, k) = k - 1, \quad f(2, k) = \text{CH}(k - 1, k - 1). \quad (2.6)$$

Proof. A family of distinct singletons contains a k -sunflower exactly when it has at least k members. This gives the first equality.

View a 2-uniform family as a graph. A sunflower of distinct edges has empty core or a one-point core. These two cases are a matching and a set of edges incident with one vertex. The graph is therefore k -sunflower-free exactly when its matching number and maximum degree are both at most $k - 1$. Equation (2.5) gives the second equality. $\square$

2.4 Products

Abbott proved the following product inequality [1]. Frankl and Wang later proved a vector-valued generalization [13].

Let X and Y be disjoint. For $\mathcal{A} \subseteq \binom{X}{u}$ and $\mathcal{B} \subseteq \binom{Y}{v}$, define

$$\mathcal{A} * \mathcal{B} = \{A \cup B : A \in \mathcal{A}, B \in \mathcal{B}\}.$$

This family is $(u + v)$ -uniform and has size $|\mathcal{A}||\mathcal{B}|$.

Lemma 2.4 (Product construction). *If $\mathcal{A}$ and $\mathcal{B}$ contain no k -sunflower, then $\mathcal{A} * \mathcal{B}$ contains no k -sunflower. Hence*

$$f(u + v, k) \geq f(u, k)f(v, k). \quad (2.7)$$

Proof. Suppose that $A_i \cup B_i$, for $1 \leq i \leq k$, form a sunflower in the product family. Intersecting with X and Y shows that the pairwise intersections among the A_i are equal, and the same holds among the B_i .

If two A_i are equal, their common pairwise intersection is that whole set. It is contained in every other A_j . Uniformity then implies that all A_i are equal. Otherwise the A_i are distinct and form a k -sunflower in $\mathcal{A}$. The latter case is impossible, so all A_i are equal. The same argument shows that all B_i are equal. The k product members are then equal, a contradiction.

Taking $\mathcal{A}$ and $\mathcal{B}$ to be extremal families on disjoint supports, the product family has $f(u, k)f(v, k)$ members, which proves equation (2.7). $\square$

Since $\log f(w, k)$ is superadditive, Fekete's lemma [12] and equation (2.7) give the extended limit

$$B(k) = \lim_{w \rightarrow \infty} f(w, k)^{1/w} = \sup_{w \geq 1} f(w, k)^{1/w}. \quad (2.8)$$

The value in equation (2.8) may be infinite. The sunflower conjecture states that it is finite for every fixed k .

3 A matching-constrained construction

We first construct larger sunflower-free families from smaller ones while controlling their matching numbers. We then specialize the resulting recurrence and combine it with a simple splitting argument to obtain a recurrence involving only $f(w, k)$. Finally, we solve this recurrence to derive an explicit lower bound on the exponential growth rate of $f(w, k)$. An alternative recurrence using a bounded-degree auxiliary graph is recorded in the appendix.

3.1 Block-size recurrence

The construction builds a w -uniform sunflower-free family from smaller families by adjoining either one auxiliary point or a larger auxiliary block, while preserving the required matching-number bound.

Theorem 3.1 (Block-size recurrence). *Let $k \geq 3$ and $2 \leq m \leq k - 1$. Put*

$$t = k - m + 1.$$

For every $w \geq t + 1$,

$$\psi(w, k, m) \geq (k - 1)\psi(w - 1, k, m) + \left\lfloor \frac{k - 1}{t} \right\rfloor f(w - t, k). \quad (3.1)$$

Proof. Take pairwise disjoint sets X, Y, Z , where $|Y| = k - 1$. On X , take a $(w - 1)$ -uniform family $\mathcal{A}$ with no k -sunflower such that

$$|\mathcal{A}| = \psi(w - 1, k, m), \quad \nu(\mathcal{A}) \leq m - 1.$$

Let $\mathcal{P}$ consist of $\lfloor (k - 1)/t \rfloor$ pairwise disjoint t -subsets of Y . For each $p \in \mathcal{P}$, take a $(w - t)$ -uniform family $\mathcal{E}_p$ on Z with no k -sunflower and with $|\mathcal{E}_p| = f(w - t, k)$. The families $\mathcal{E}_p$ may have the same support. Define

$$\begin{aligned} \mathcal{F} = & \{T \cup \{y\} : T \in \mathcal{A}, y \in Y\} \\ & \cup \{p \cup S : p \in \mathcal{P}, S \in \mathcal{E}_p\}. \end{aligned} \quad (3.2)$$

The members in the first and second lines meet Y in one and t points, respectively. Since $t \geq 2$, the two lines are disjoint. All members within each line are distinct. Consequently,

$$|\mathcal{F}| = |Y||\mathcal{A}| + \sum_{p \in \mathcal{P}} |\mathcal{E}_p| = (k - 1)\psi(w - 1, k, m) + \left\lfloor \frac{k - 1}{t} \right\rfloor f(w - t, k),$$

which is the right-hand side of equation (3.1).

Consider a matching in $\mathcal{F}$ with q members from the first line and b members from the second. If $b = 0$, the X -parts form a matching in $\mathcal{A}$, so $q \leq m - 1$. If $b \geq 1$, then, because the matching members are pairwise disjoint, their intersections with Y are also pairwise disjoint, and

$$q + tb \leq k - 1.$$

It follows that

$$q + b \leq k - 1 - (t - 1)b \leq k - t = m - 1.$$

Hence $\nu(\mathcal{F}) \leq m - 1$.

Suppose that $F_1, \dots, F_k$ form a sunflower in $\mathcal{F}$. Every F_i meets the $(k-1)$ -point set Y . By the pigeonhole principle, some $y \in Y$ therefore belongs to at least two of them. By lemma 2.1, it belongs to all of them.

If all F_i come from the first line of equation (3.2), write $F_i = T_i \cup \{y\}$. Deleting y gives a k -sunflower of distinct members of $\mathcal{A}$.

If all F_i come from the second line, every corresponding block in $\mathcal{P}$ contains y . The blocks in $\mathcal{P}$ are disjoint, so all F_i use the same block p . Deleting p gives a k -sunflower of distinct members of $\mathcal{E}_p$.

It remains to consider a sunflower using both lines. A member from the first line and a member from the second intersect exactly in $\{y\}$. The core is therefore $\{y\}$. The X -parts of the members from the first line form a matching in $\mathcal{A}$, so there are at most $m-1$ such members. All members from the second line contain y . Since the blocks in $\mathcal{P}$ are disjoint, they all use the same block p . The set $p \setminus \{y\}$ is nonempty because $t \geq 2$, so there is at most one member from the second line. The sunflower has at most $m \leq k-1$ members, a contradiction. $\square$

The case $m = k-1$ gives the recurrence used for the exponential estimate.

Corollary 3.2. *For every $k \geq 3$ and $w \geq 3$,*

$$\psi(w, k, k-1) \geq (k-1)\psi(w-1, k, k-1) + \left\lfloor \frac{k-1}{2} \right\rfloor f(w-2, k). \quad (3.3)$$

3.2 Unrestricted families

The following inequalities connect the matching-constrained quantities to $f(w, k)$. The second is a case of the Abbott–Hanson disjoint-support split [4].

Lemma 3.3 (Star and split). *For every $k \geq 3$ and $w \geq 2$,*

$$\psi(w, k, 2) \geq f(w-1, k), \quad (3.4)$$

$$f(w, k) \geq \psi(w, k, k-1) + \psi(w, k, 2). \quad (3.5)$$

Proof. For equation (3.4), take an extremal $(w-1)$ -uniform family with $f(w-1, k)$ members and add one new point to every member. The resulting family has the same size and is intersecting. Deleting the new point from a sunflower would give a sunflower in the original family, so the resulting family is counted by $\psi(w, k, 2)$. This proves equation (3.4).

For equation (3.5), take extremal families attaining $\psi(w, k, k-1)$ and $\psi(w, k, 2)$ and place them on disjoint supports. Their union has size $\psi(w, k, k-1) + \psi(w, k, 2)$. A sunflower with nonempty core cannot use members from both supports. A sunflower with empty core would be a matching of size k , but the matching number of the union is at most

$$(k-2) + 1 = k-1.$$

Thus the union contains no k -sunflower, which proves equation (3.5). $\square$

For $k \geq 4$, Frankl and Wang proved

$$\psi(3, k, 2) = f(2, k) \quad (3.6)$$

[13, Theorem 1.15]. Thus the star bound equation (3.4) is tight when $w = 3$.

3.3 Exponential rate

Combining corollary 3.2 and lemma 3.3 yields simultaneous recursive lower bounds for $\psi(w, k, k-1)$ and $f(w, k)$. We use a_w to track lower bounds for $\psi(w, k, k-1)$ and b_w to track lower bounds for $f(w, k)$.

Proposition 3.4 (Recursive lower bounds). *Fix $k \geq 3$. Let a_3, b_2, b_3 be positive integers satisfying*

$$a_3 \leq \psi(3, k, k-1), \quad b_2 \leq f(2, k), \quad b_3 \leq f(3, k).$$

For $w \geq 4$, define

$$a_w = (k-1)a_{w-1} + \left\lfloor \frac{k-1}{2} \right\rfloor b_{w-2}, \quad (3.7)$$

$$b_w = a_w + b_{w-1}. \quad (3.8)$$

Then

$$a_w \leq \psi(w, k, k-1), \quad b_w \leq f(w, k) \quad (w \geq 4).$$

Proof. We prove both inequalities simultaneously by induction. For $w \geq 4$, assume that

$$a_{w-1} \leq \psi(w-1, k, k-1), \quad b_{w-2} \leq f(w-2, k), \quad b_{w-1} \leq f(w-1, k).$$

When $w = 4$, these are precisely the three assumed initial bounds. By equation (3.3),

$$\begin{aligned} \psi(w, k, k-1) &\geq (k-1)\psi(w-1, k, k-1) + \left\lfloor \frac{k-1}{2} \right\rfloor f(w-2, k) \\ &\geq (k-1)a_{w-1} + \left\lfloor \frac{k-1}{2} \right\rfloor b_{w-2} = a_w. \end{aligned}$$

Using equations (3.4) and (3.5), we then obtain

$$\begin{aligned} f(w, k) &\geq \psi(w, k, k-1) + \psi(w, k, 2) \\ &\geq a_w + f(w-1, k) \\ &\geq a_w + b_{w-1} = b_w. \end{aligned}$$

These two bounds provide the induction hypotheses at the next index. $\square$

The two recurrences reduce to one recurrence for b_w . For $w \geq 5$, equation (3.8) at index $w-1$ gives

$$a_{w-1} = b_{w-1} - b_{w-2}.$$

Substitution into equations (3.7) and (3.8) gives

$$b_w = kb_{w-1} - \left\lfloor \frac{k-1}{2} \right\rfloor b_{w-2}, \quad w \geq 5. \quad (3.9)$$

This recurrence starts at $w = 5$ because b_3 need not equal $a_3 + b_2$. The construction first gives a_4 and then sets $b_4 = a_4 + b_3$.

Theorem 3.5 (Exponential rate). *For every $k \geq 3$,*

$$B(k) \geq r_k, \quad r_k = \begin{cases} \frac{k + \sqrt{k^2 - 2k}}{2}, & k \text{ even}, \\ \frac{k + \sqrt{k^2 - 2k + 2}}{2}, & k \text{ odd}. \end{cases} \quad (3.10)$$

Proof. Use the sequences in proposition 3.4. Consider the polynomial

$$x^2 - kx + c, \quad c = \left\lceil \frac{k-1}{2} \right\rceil.$$

It is positive at 0, negative at 1, and tends to infinity with x . Its roots $s < r$ therefore satisfy $0 < s < 1 < r$. Put

$$u_w = b_w - sb_{w-1}.$$

Equation (3.9) and the identities $r + s = k$ and $rs = c$ give

$$\begin{aligned} u_w &= b_w - sb_{w-1} \\ &= (k-s)b_{w-1} - cb_{w-2} \\ &= rb_{w-1} - rsb_{w-2} \\ &= r(b_{w-1} - sb_{w-2}) = ru_{w-1} \quad (w \geq 5). \end{aligned}$$

The initial value is

$$u_4 = b_4 - sb_3 = a_4 + (1-s)b_3 > 0.$$

Therefore

$$u_w = u_4 r^{w-4} \quad (w \geq 4).$$

Since $b_w = u_w + sb_{w-1} \geq u_w$ and $b_w \leq f(w, k)$,

$$f(w, k)^{1/w} \geq u_4^{1/w} r^{(w-4)/w}.$$

Letting w tend to infinity gives $B(k) \geq r$. Substituting the two values of c into the quadratic formula gives equation (3.10). $\square$

Corollary 3.6. *The bound in theorem 3.5 gives*

$$B(4) \geq 2 + \sqrt{2}, \quad B(5) \geq \frac{5 + \sqrt{17}}{2}, \quad B(6) \geq 3 + \sqrt{6}.$$

As k tends to infinity,

$$r_k = \begin{cases} k - \frac{1}{2} - \frac{1}{4k} + O(k^{-2}), & k \text{ even}, \\ k - \frac{1}{2} + \frac{1}{4k} + O(k^{-2}), & k \text{ odd}. \end{cases} \quad (3.11)$$

Proof. The three displayed values follow by substitution. For the asymptotic expansion, use

$$\begin{aligned} \sqrt{k^2 - 2k} &= k - 1 - \frac{1}{2k} + O(k^{-2}), \\ \sqrt{k^2 - 2k + 2} &= k - 1 + \frac{1}{2k} + O(k^{-2}), \end{aligned}$$

and substitute into equation (3.10). $\square$

For comparison, Abbott, Hanson, and Sauer proved $B(3) \geq \sqrt{10}$ [3]. This is stronger than the $k = 3$ specialization of theorem 3.5. Their stronger three-petal estimate comes from a substitution construction whose proof is specific to three petals, whereas theorem 3.5 applies to every $k \geq 3$. The constructions of Abbott and Exoo imply

$$B(4) \geq 38^{1/3}, \quad B(5) \geq \sqrt{20}, \quad B(6) \geq 146^{1/3}$$

[5]. The three corresponding bounds in corollary 3.6 are larger.

The exponential-rate argument uses equation (3.3). The other matching caps in theorem 3.1 give additional finite recurrences. Appendix A records the alternative capacity recurrence.

4 Upper bounds for triples

This section proves an upper bound for $f(3, k)$ that applies for every $k \geq 4$ and has leading term $2k^3$. At $k = 5, 6, 7$ it gives

$$f(3, 5) \leq 146, \quad f(3, 6) \leq 255, \quad f(3, 7) \leq 474.$$

The argument begins with a largest collection of pairwise disjoint triples. Every member of the family meets their union, since otherwise the collection could be enlarged. We classify the members by the number of points they have in this union. Members meeting it in one point give graphs on the remaining points; maximality forces certain pairs of these graphs to be cross-intersecting, while the absence of sunflowers bounds their degrees and matching numbers. Combining these graph bounds with an incidence count gives the general upper bound.

The maximum-matching framework is inherited from Abbott and Hanson: they decompose a family around disjoint triples and analyze the resulting residue graphs, obtaining the bound $3(k-1)$ in one case and $4(k-1)$ in general [4]. Our graph lemma sharpens the general term to $3(k-1)$, and the subsequent argument replaces the residue matching cap $k-1$ by $k-2$. Frankl and Wang give another general upper bound for triple families [13, Theorem 1.14]. Both comparison bounds apply only when $k \geq 7$; asymptotically, the Frankl–Wang bound has the smaller leading term $\frac{5}{3}k^3$.

4.1 Cross-intersecting graphs

We first establish the required bound for three pairwise cross-intersecting graphs.

Lemma 4.1 (Cross-intersecting graph bound). *Let A, B, C be simple graphs whose edge families are pairwise cross-intersecting. Suppose that each graph has maximum degree at most $k-1$, where $k \geq 4$. If at least two of the graphs are nonempty, then*

$$|E(A)| + |E(B)| + |E(C)| \leq 3(k-1).$$

Proof. First suppose that none of A, B, C contains two disjoint edges; equivalently, each nonempty graph is intersecting. An intersecting graph is a star or a subgraph of a triangle. Indeed, if it contains the edges xy and xz , then any edge avoiding x must be yz . If such an edge occurs, no further edge is possible outside this triangle.

If one of the graphs contains three edges of a star centered at x , every edge in all three graphs contains x . A two-point edge avoiding x cannot meet three distinct leaves. The degree bound gives at most $k-1$ edges in each graph. If one graph contains a triangle, every edge in all three graphs is an edge of that triangle, and the total is at most $9 \leq 3(k-1)$. In the remaining case each graph has at most two edges.

Now suppose that one graph, say A , contains disjoint edges $P = \{p, q\}$ and $Q = \{r, s\}$. Every edge of B and C has one endpoint in P and one in Q . Thus both are subgraphs of

$$K = \{pr, ps, qr, qs\}.$$

Assume that exactly one of B, C is nonempty, say B . The possible edge sets of B , up to symmetry, give the following bounds:

$E(B)$	one	two adjacent	two opposite	three	four
$ E(A) $	$2(k-1)$	k	4	3	2
$ E(A) + E(B) $	$2k-1$	$k+2$	6	6	6.

For one prescribed edge, every edge of A belongs to the union of its two endpoint-stars. For two adjacent prescribed edges, every edge of A belongs to their common-endpoint star or is the edge joining their other endpoints. The remaining three bounds follow by listing the two-point sets that meet every prescribed edge. Each total in the last row is at most $3(k-1)$.

Finally, suppose that both B and C are nonempty. Each edge of K is disjoint only from its opposite edge. Cross-intersection therefore gives

$$|E(B)| + |E(C)| \leq 4. \quad (4.1)$$

Put $U = E(B) \cup E(C)$. If $|U| = 1$, then B and C consist of the same single edge. Every edge of A meets this edge, so

$$|E(A)| + |E(B)| + |E(C)| \leq 2(k-1) + 2 = 2k \leq 3(k-1).$$

If $|U| \geq 2$, the possibilities for U are two adjacent edges, two opposite edges, three edges, or all four edges of K . The corresponding bounds for $|E(A)|$ are $k, 4, 3, 2$. Together with equation (4.1), the total is at most

$$\max\{k+4, 8\} \leq 3(k-1).$$

□

4.2 Maximum-matching bound

Theorem 4.2. *For every $k \geq 4$, put*

$$h_k = \text{CH}(k-1, k-1) = f(2, k), \quad c_k = \max\{\text{CH}(k-1, k-2), 3(k-1)\}.$$

Then

$$f(3, k) \leq \frac{k-1}{2}(c_k + 3h_k - 1). \quad (4.2)$$

Proof. Let $\mathcal{F}$ be a 3-uniform family without a k -sunflower such that $|\mathcal{F}| = f(3, k)$. Choose a maximum matching $M_1, \dots, M_v$ and put

$$T = M_1 \cup \dots \cup M_v.$$

Equation (2.2) gives $v \leq k-1$. Every member of $\mathcal{F}$ meets T , since a member disjoint from T would extend the matching.

For $j \in \{1, 2, 3\}$, let

$$t_j = |\{F \in \mathcal{F} : |F \cap T| = j\}|.$$

By lemmas 2.2 and 2.3, every point has degree at most h_k . Counting incidences between the members of $\mathcal{F}$ and the $3v$ points of T gives

$$t_1 + 2t_2 + 3t_3 \leq 3vh_k. \quad (4.3)$$

The matching members themselves give

$$t_3 \geq v. \quad (4.4)$$

Fix i and write $M_i = \{a, b, c\}$. On the points outside T , define

$$S_a = \{F \setminus \{a\} : F \in \mathcal{F}, F \cap T = \{a\}\},$$

and define S_b, S_c in the same way. These graphs are pairwise cross-intersecting. If an edge of S_a and an edge of S_b were disjoint, their corresponding triples, together with the members M_j for $j \neq i$, would form a matching of size $v + 1$.

Each residue graph has maximum degree at most $k - 1$. Indeed, if a point x outside T were incident with k edges $\{x, z_1\}, \dots, \{x, z_k\}$ in S_a , then the corresponding triples $\{a, x, z_1\}, \dots, \{a, x, z_k\}$ would form a k -sunflower with core $\{a, x\}$. Moreover, the link $\text{Lk}_{\mathcal{F}}(a)$ contains the edge $\{b, c\}$, which is disjoint from every edge of S_a . Thus a matching of size $k - 1$ in S_a , together with $\{b, c\}$, would form a matching of size k in the link and hence a k -sunflower with core $\{a\}$. Therefore

$$\nu(S_a) \leq k - 2, \quad |E(S_a)| \leq \text{CH}(k - 1, k - 2),$$

and the same inequalities hold for S_b and S_c .

If at most one residue graph is nonempty, their combined size is at most $\text{CH}(k - 1, k - 2)$. If at least two are nonempty, lemma 4.1 bounds their combined size by $3(k - 1)$. Consequently, the members counted by t_1 that meet a fixed M_i contribute at most c_k . Every member counted by t_1 meets a unique matching triple M_i in exactly one point and is represented by exactly one edge in one of its three residue graphs. Summing over the v matching triples therefore gives

$$t_1 \leq vc_k. \tag{4.5}$$

Eliminating t_2 from equation (4.3) and using equations (4.4) and (4.5), we obtain

$$\begin{aligned} |\mathcal{F}| &= t_1 + t_2 + t_3 \\ &\leq \frac{3vh_k + t_1 - t_3}{2} \\ &\leq \frac{v}{2}(c_k + 3h_k - 1) \\ &\leq \frac{k - 1}{2}(c_k + 3h_k - 1). \end{aligned}$$

□

4.3 Numerical consequences

For $k \geq 5$, one has $\text{CH}(k - 1, k - 2) \geq (k - 1)(k - 2) \geq 3(k - 1)$, so $c_k = \text{CH}(k - 1, k - 2)$. Lemma 4.1 controls the case of two or three nonempty residue graphs. At $k = 4$, it gives $c_4 = 9$, and theorem 4.2 yields $f(3, 4) \leq 57$.

Corollary 4.3. *The general bound gives*

$$f(3, 5) \leq 146, \quad f(3, 6) \leq 255, \quad f(3, 7) \leq 474.$$

Proof. Equation (2.5) gives

k	h_k	$\text{CH}(k - 1, k - 2)$	c_k	$\frac{k - 1}{2}(c_k + 3h_k - 1)$
5	20	14	14	146
6	27	22	22	255
7	42	33	33	474.

Apply theorem 4.2.

□

5 Bounds for $f(3, 4)$

This section proves $39 \leq f(3, 4) \leq 49$. The lower bound is given by an explicit family of 39 triples. For the upper bound, we assume that a 50-member family without a 4-sunflower exists and choose a maximum matching of three triples. We analyze the local structure around these triples, decompose the family into rooted blocks and cross classes, and reduce the possible configurations to four numerical profiles. Finite exhaustive searches then rule out all four profiles.

5.1 A family of 39 triples

Theorem 5.1. *One has*

$$f(3, 4) \geq 39.$$

Proof. Let V_1, V_2, V_3 be pairwise disjoint sets of size 3. Take four further points $a, \ell_{12}, \ell_{13}, \ell_{23}$. Define

$$\begin{aligned}\mathcal{G} &= \{V_1, V_2, V_3\}, \\ \mathcal{P} &= \{\{a\} \cup P : P \in \binom{V_i}{2}, 1 \leq i \leq 3\}, \\ \mathcal{Q} &= \{\{\ell_{ij}, x, y\} : 1 \leq i < j \leq 3, x \in V_i, y \in V_j\}.\end{aligned}$$

Put $\mathcal{F}_{39} = \mathcal{G} \cup \mathcal{P} \cup \mathcal{Q}$. These three families are disjoint and

$$|\mathcal{F}_{39}| = 3 + 3 \binom{3}{2} + 3 \cdot 3^2 = 39.$$

We exclude the three possible core sizes. Every pair has degree at most 3. The pairs $\{\ell_{ij}, x\}$ with $x \in V_i \cup V_j$ have degree 3; all other pair types have degree at most 2. Thus there is no 4-sunflower with a two-point core.

The link at a is the disjoint union of three triangles. The link at ℓ_{ij} is the complete bipartite graph between V_i and V_j . For $x \in V_i$, its link is the disjoint union of a triangle on $\{a\} \cup (V_i \setminus \{x\})$ and two three-edge stars. The stars have centres ℓ_{ij} and leaves V_j , for the two indices $j \neq i$. Each of these link graphs has matching number 3. Hence there is no 4-sunflower with a one-point core.

It remains to exclude four disjoint members. Let g be the number of group triples V_i in a matching. If $g = 3$, no further member is disjoint from all three groups. If $g = 2$, every remaining member uses points of the third group. The only possibilities are the third group triple and the three members of $\mathcal{P}$ supported on that group. These four members are pairwise intersecting, so at most one can be added.

Suppose that $g = 1$, say V_1 is used. Among members avoiding V_1 , the only special points are a and ℓ_{23} , apart from the two group triples V_2, V_3 . If one of those group triples is used, at most one further member can be disjoint from it. If neither is used, at most two members can be chosen because every available member contains a or ℓ_{23} . Thus the matching has at most three members.

Finally suppose that $g = 0$ and that four non-group members are disjoint. They must use the four special points $a, \ell_{12}, \ell_{13}, \ell_{23}$ exactly once. The member containing a uses two points of some group, say V_1 . The members containing ℓ_{12} and ℓ_{13} use two further distinct points of V_1 . This would require four points in V_1 , a contradiction. Hence $\nu(\mathcal{F}_{39}) \leq 3$, so no 4-sunflower has empty core. $\square$

5.2 Local structure around a triple

The upper bound follows from finite enumeration. The next lemmas give the mathematical reduction and define the objects that are enumerated.

Lemma 5.2 (Degree bounds). *Let $\mathcal{F}$ be a 3-uniform family without a 4-sunflower. Every pair has degree at most 3, and every point has degree at most 10.*

Proof. Four distinct triples containing the same pair form a sunflower with that pair as core. For a point x , the link graph $\text{Lk}_{\mathcal{F}}(x)$ has maximum degree and matching number at most 3. Equation (2.5) gives

$$d_{\mathcal{F}}(x) = |E(\text{Lk}_{\mathcal{F}}(x))| \leq \text{CH}(3, 3) = 10.$$

□

Fix $S = \{s_1, s_2, s_3\} \in \mathcal{F}$. For $s \in S$, define the exact-singleton petal graph

$$L'_s(S) = \{F \setminus \{s\} : F \in \mathcal{F}, F \cap S = \{s\}\}.$$

Lemma 5.3 (Exact-singleton links). *For every $s \in S$,*

$$\Delta(L'_s(S)) \leq 3, \quad \nu(L'_s(S)) \leq 2, \quad |E(L'_s(S))| \leq 7.$$

Proof. The degree bound follows from lemma 5.2. Three disjoint edges of $L'_s(S)$, together with the edge $S \setminus \{s\}$ in $\text{Lk}_{\mathcal{F}}(s)$, would give a four-matching in this link. Thus $\nu(L'_s(S)) \leq 2$, and equation (2.5) gives $\text{CH}(3, 2) = 7$. □

In the maximum-matching decomposition used below, the members associated exclusively with one matching triple form an intersecting family containing that triple; the following definition records this structure.

A rooted block is an intersecting, 3-uniform family $\mathcal{A}$ without a 4-sunflower, together with a distinguished member $T \in \mathcal{A}$. Put

$$p(\mathcal{A}, T) = |\{A \in \mathcal{A} \setminus \{T\} : |A \cap T| = 2\}|.$$

Lemma 5.4 (Rooted blocks). *Every intersecting, 3-uniform family without a 4-sunflower has at most 10 members [13, Proposition 6.3]. The rooted isomorphism-class counts used below are*

rooted block list	classes
$ \mathcal{A} = 8$ and $p(\mathcal{A}, T) = 0$	45
$ \mathcal{A} = 9$ and $p(\mathcal{A}, T) = 0$	27
all rooted blocks with $ \mathcal{A} = 9$	665
all rooted blocks with $ \mathcal{A} = 10$	114

Computer-assisted proof. The cited result gives the size bound. The enumeration below independently recovers it and establishes the rooted class counts. Relabel the anchor as $\{0, 1, 2\}$. Adjoin the other members in lexicographic order. Outside labels obey the first-appearance rule: a new outside label is the least unused label. This preserves one representative of every rooted isomorphism type. To see this, choose a labelling with least sorted member list. If a larger label appeared before a smaller unused label, interchanging the two would decrease the list.

A block with m members uses at most $2(m-1)$ outside points. The enumeration at sizes 8, 9, 10, 11 therefore needs at most 14, 16, 18, 20 outside labels, respectively. The nodes of the enumeration tree are partial rooted blocks. A node is rejected exactly when it contains a repeated triple, a disjoint pair, or a 4-sunflower. Canonicalization permits every anchor permutation and every permutation of the outside support, and gives the four class counts in the statement. No valid node reaches size 11, while valid size-10 blocks exist. $\square$

Define the closed meeting neighborhood of S by

$$\mathcal{N}_{\mathcal{F}}[S] = \{F \in \mathcal{F} : F \cap S \neq \emptyset\}.$$

For $e \in \binom{S}{2}$, let n_e be the number of members other than S containing e . Put

$$s = \sum_{e \in \binom{S}{2}} n_e, \quad \mu_x = \sum_{\substack{e \in \binom{S}{2} \\ x \in e}} n_e \quad (x \in S).$$

By lemmas 5.2 and 5.3,

$$0 \leq n_e \leq 2, \quad |E(L'_x(S))| \leq \min\{7, 9 - \mu_x\}.$$

Each member meeting S in two points contributes to μ_x at both of those points, so

$$\sum_{x \in S} \mu_x = 2s.$$

Consequently,

$$\begin{aligned} |\mathcal{N}_{\mathcal{F}}[S]| &= 1 + s + \sum_{x \in S} |E(L'_x(S))| \\ &\leq 1 + s + \min\{21, 27 - 2s\} \leq 25. \end{aligned} \tag{5.1}$$

Lemma 5.5 (Neighborhood equality). *Equality in equation (5.1) is possible only when*

$$n_e = 1 \quad (e \in \binom{S}{2}), \quad |E(L'_x(S))| = 7, \quad d_{\mathcal{F}}(x) = 10 \quad (x \in S).$$

The three members meeting S in two points then have a common third point y , and y is outside every graph $L'_x(S)$.

Proof. For integral $0 \leq s \leq 6$, the last expression in equation (5.1) is uniquely maximized at $s = 3$. Equality forces all three singleton links to have seven edges and all μ_x to equal 2. The three equations for the n_e give $n_e = 1$ for every internal pair.

Up to isomorphism, the unique seven-edge graph with maximum degree at most 3 and matching number at most 2 is

$$R = K_5 - (P_3 \cup K_2).$$

Its degree sequence is $(3, 3, 3, 3, 2)$, and $R - v$ has a two-matching for every vertex v . This follows by listing the edges incident with the four endpoints of a maximum two-matching and applying the degree and matching bounds.

Write the three two-point members as

$$\{s_2, s_3, y_1\}, \quad \{s_1, s_3, y_2\}, \quad \{s_1, s_2, y_3\}.$$

If $y_2 \neq y_3$, the two corresponding edges in $\text{Lk}_{\mathcal{F}}(s_1)$ are disjoint. Deleting y_2, y_3 from $L'_{s_1}(S) \cong R$ must then destroy every two-matching. If, say, y_2 were not a vertex of R , deleting

y_2, y_3 would remove at most one vertex of R , and the remaining graph would still contain a two-matching. Hence both y_2 and y_3 belong to R . Since each already occurs with s_1 in one of the two displayed members, the pair-degree bound gives degree at most two in R . They must therefore both be the unique degree-two vertex of R , a contradiction. Hence $y_2 = y_3$, and symmetry gives $y_1 = y_2 = y_3 =: y$.

The pair $\{s_1, y\}$ already occurs in two displayed members. If y were a vertex of $L'_{s_1}(S)$, its degree there would be at most 1, whereas R has minimum degree 2. Thus $y \notin V(L'_{s_1}(S))$, and the same holds at the other two points of S . $\square$

5.3 The maximum-matching decomposition

Lemma 5.6. *If $\mathcal{F}$ is 3-uniform, has no 4-sunflower, and $\nu(\mathcal{F}) \leq 2$, then $|\mathcal{F}| \leq 35$. Equivalently,*

$$\psi(3, 4, 3) \leq 35.$$

Proof. The case $\nu(\mathcal{F}) \leq 1$ follows from lemma 5.4. Otherwise fix a maximum matching T_1, T_2 . Apart from the anchors, classify a member by its intersection sizes with (T_1, T_2) . Let m, p, q, r count, in both orientations, the types

$$(2, 1), \quad (2, 0), \quad (1, 1), \quad (1, 0),$$

respectively. These types exhaust the family, so

$$|\mathcal{F}| = 2 + m + p + q + r.$$

Summing degrees over the six anchor points gives

$$3m + 2p + 2q + r \leq 54. \tag{5.2}$$

Indeed, the six points have total degree at most 60, while the two anchors themselves contribute six incidences. Among the other members, the four types contribute 3, 2, 2, 1 anchor incidences, respectively. Summing the exact-singleton link bound over the same points gives

$$m + 2q + r \leq 42. \tag{5.3}$$

There are six such links, each with at most seven edges; a member of type m, q, r is counted once, twice, or once, respectively, while a member of type p is not counted. For $i = 1, 2$, the members missing the other anchor form an intersecting rooted block. Applying lemma 5.4 to both blocks gives

$$p + r \leq 18. \tag{5.4}$$

Each block has at most ten members and includes its anchor, so the two blocks contain at most eighteen non-anchor members in total. One quarter of each of the first two inequalities and one half of the third give

$$m + p + q + r \leq \frac{54}{4} + \frac{42}{4} + \frac{18}{2} = 33.$$

Thus $|\mathcal{F}| \leq 35$. $\square$

Assume for a contradiction that $\mathcal{F}$ has $N = 50$ members and no 4-sunflower. A matching of four triples would be a 4-sunflower with empty core, so $\nu(\mathcal{F}) \leq 3$. By lemma 5.6, the

inequality $\nu(\mathcal{F}) \leq 2$ would imply $|\mathcal{F}| \leq 35$. Thus $\nu(\mathcal{F}) = 3$. Fix a maximum matching T_1, T_2, T_3 . For $\{i, j, k\} = \{1, 2, 3\}$, define

$$\begin{aligned}\mathcal{F}_i &= \{F \in \mathcal{F} : F \cap T_i \neq \emptyset, F \cap T_j = F \cap T_k = \emptyset\}, & a_i &= |\mathcal{F}_i|, \\ \mathcal{C}_i &= \{F \in \mathcal{F} : F \cap T_i = \emptyset, F \cap T_j \neq \emptyset, F \cap T_k \neq \emptyset\}, & c_i &= |\mathcal{C}_i|, \\ \mathcal{Z} &= \{F \in \mathcal{F} : F \cap T_1, F \cap T_2, F \cap T_3 \text{ are all nonempty}\}, & z &= |\mathcal{Z}|.\end{aligned}$$

These classes partition $\mathcal{F}$ and

$$N = \sum_{i=1}^3 a_i + \sum_{i=1}^3 c_i + z. \quad (5.5)$$

Each $\mathcal{F}_i$ is an intersecting rooted block: two disjoint members, together with T_j, T_k , would form a four-matching. Hence

$$1 \leq a_i \leq 10. \quad (5.6)$$

Let $\mathcal{D}_i$ be the family of members disjoint from T_i , and put

$$D_i = |\mathcal{D}_i|, \quad N_i = |\mathcal{N}_{\mathcal{F}}[T_i]|, \quad V_i = N - 2 + a_i - D_i.$$

This choice connects the local neighborhood and block bounds to the global identity (5.8) below, which restricts the possible 50-member profiles. The definitions give

$$D_i = a_j + a_k + c_i, \quad N_i = N - D_i, \quad V_i = N_i + a_i - 2. \quad (5.7)$$

Each $\mathcal{F}_i$ is counted in two of the families $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$, each $\mathcal{C}_i$ is counted in one, and $\mathcal{Z}$ is counted in none. Therefore

$$\sum_{i=1}^3 D_i = 2 \sum_{i=1}^3 a_i + \sum_{i=1}^3 c_i.$$

Combining this identity with equations (5.5) and (5.7) gives

$$\sum_{i=1}^3 V_i = 2N - 6 + z. \quad (5.8)$$

The family $\mathcal{D}_i$ has matching number at most 2. Therefore lemma 5.6 and equations (5.1), (5.6) and (5.7) give

$$D_i \leq 35, \quad N_i \leq 25, \quad V_i \leq 33. \quad (5.9)$$

5.4 Rigid blocks and finite profiles

Call $\mathcal{F}_i$ rigid when $N_i = 25$. By lemma 5.5, the three members meeting T_i in two points then have a common third point y .

Lemma 5.7 (Rigid-block interaction). *Let $\mathcal{F}_i$ be rigid.*

- (i) *If y lies outside $T_1 \cup T_2 \cup T_3$, then $a_i = 4$.*
- (ii) *If $a_i \geq 5$, then y lies in another anchor, say T_j . No non-anchor member of $\mathcal{F}_i$ meets T_i in two points. Thus $p(\mathcal{F}_i, T_i) = 0$, and the $a_i - 1$ non-anchor members meet T_i in exactly one point.*

(iii) Suppose that $y \in T_j$, put

$$\mathcal{B}_y = \{F \in \mathcal{F}_j : y \notin F\}, \quad r = a_i - 1.$$

For the three values needed below,

$$\begin{array}{c|ccc} r & 7 & 8 & 9 \\ \hline |\mathcal{B}_y| & \leq 3 & \leq 2 & \leq 1. \end{array}$$

If $\mathcal{F}_j$ is not rigid, then $d_{\mathcal{F}_j}(y) \leq 7$. When $r = 8$, a non-rigid $\mathcal{F}_j$ cannot have size 9. If $\mathcal{F}_j$ is rigid, then either $a_j = 4$ or $d_{\mathcal{F}_j}(y) \leq 2$.

Proof (partly computer-assisted). Suppose first that y is outside the three anchors. A member of $\mathcal{F}_i$ meeting T_i in one point must intersect the two-point member supported on the other two points of T_i . It can do so only through y , contrary to lemma 5.5. The anchor and the three two-point members are therefore all of $\mathcal{F}_i$. This proves (i). If $a_i \geq 5$, part (i) and $y \notin T_i$ place y in another anchor. The three two-point members then lie outside $\mathcal{F}_i$, which proves (ii).

For (iii), delete the anchor point from every one-point member of $\mathcal{F}_i$. This gives three edge families R_1, R_2, R_3 , indexed by the points of T_i . They are pairwise cross-intersecting, each has maximum degree at most 3 and matching number at most 2, and their total size is r .

Every member of $\mathcal{B}_y$ must cover each edge in $R_1 \cup R_2 \cup R_3$ with its one- or two-point part outside the anchors. Indeed, let $A = \{s\} \cup e \in \mathcal{F}_i$ correspond to an edge $e \in R_s$. If $B \in \mathcal{B}_y$ avoided e , then A, B, T_k , and the rigid two-point member $(T_i \setminus \{s\}) \cup \{y\}$ would be four disjoint triples, where $\{i, j, k\} = \{1, 2, 3\}$.

It remains to justify the finite cover bounds. Let R_1 be largest, so $|E(R_1)| \geq 3$. If R_1 is a three-edge star, cross-intersection forces the whole union to be a star. Every one- or two-point cover contains its centre. All r one-point members of $\mathcal{F}_i$ and all members of $\mathcal{B}_y$ then contain that point, so the point-degree bound gives $|\mathcal{B}_y| \leq 10 - r$.

If R_1 is a triangle, every cross-intersecting edge lies on its three vertices. Otherwise R_1 has a two-matching, and every edge in the other two graphs lies in $V(R_1)$. By definition, $V(R_1)$ contains only vertices incident with an edge. If c is the number of connected components of R_1 , then $|V(R_1)| \leq |E(R_1)| + c$. Choosing one edge from each component gives a matching, so $c \leq \nu(R_1)$. Therefore, for $|E(R_1)| \leq 6$,

$$|V(R_1)| \leq |E(R_1)| + \nu(R_1) \leq 8.$$

The unique seven-edge graph from lemma 5.5 has five vertices. Thus every non-star case has a representative on eight vertices. The cover enumerator exhausts the three edge sets and their one- and two-point covers on these eight vertices. It applies the pair-degree and point-degree bounds and the necessary matching tests in the links of the outside points. The maxima are 3, 2, 1 for $r = 7, 8, 9$. The calculation uses only necessary conditions, so these are upper bounds for the original family.

To bound $d_{\mathcal{F}_j}(y)$, observe that the three rigid two-point members give a triangle in $L'_y(T_j)$. The one-point members of $\mathcal{F}_j$ through y have petals disjoint from this triangle. Let p_y and r_y be the numbers of two-point and one-point members of $\mathcal{F}_j$ through y . The degree and matching bounds give

$$p_y \leq \mu_y, \quad |E(L'_y(T_j))| \geq 3 + r_y.$$

Since $1 + \mu_y + |E(L'_y(T_j))| \leq 10$, we obtain $p_y + r_y \leq 6$ and therefore

$$d_{\mathcal{F}_j}(y) = 1 + p_y + r_y \leq 7.$$

Suppose that $r = 8$, the family $\mathcal{F}_j$ is non-rigid, and $a_j = 9$. The cover and degree bounds force

$$|\mathcal{B}_y| = 2, \quad p_y + r_y = 6, \quad p_y = \mu_y, \quad |E(L'_y(T_j))| = 3 + r_y.$$

The link at y already contains the two disjoint edges $T_j \setminus \{y\}$ and any edge of the rigid triangle on T_i . Every one-point petal in $\mathcal{F}_j$ avoids both T_i and $T_j \setminus \{y\}$. Two disjoint such petals would therefore complete a four-matching in the link at y . Thus the r_y one-point petals in $\mathcal{F}_j$ are intersecting, so $r_y \leq 3$, while $\mu_y \leq 4$. Hence (p_y, r_y) is $(3, 3)$ or $(4, 2)$.

Write $T_j = \{y, b, b'\}$. For $B \in \mathcal{B}_y$, call $B \setminus T_j$ its outside part. Each outside part has size one or two and covers $R_1 \cup R_2 \cup R_3$. If both have size two, they are disjoint, equal, or intersect in exactly one point. Together with the singleton case, these are the four cases treated below.

Case 1: one outside part is a singleton $\{x\}$. All eight petals from $\mathcal{F}_i$ contain x . Intersectingness of $\mathcal{F}_j$ also puts x in all $r_y \geq 2$ one-point petals through y . The member of $\mathcal{B}_y$ contains x as well, so $d_{\mathcal{F}}(x) \geq 11$.

Case 2: the outside parts are disjoint pairs C, D . If the two members of $\mathcal{B}_y$ use different anchor points, then the two full triples are disjoint, contrary to the intersectingness of $\mathcal{F}_j$. Suppose instead that they use the same anchor point, say b . A two-point member of $\mathcal{F}_j$ through y that contains b' would need its third point to belong to both C and D in order to meet both members of $\mathcal{B}_y$, which is impossible. Thus every two-point member through y contains b . The pair $\{y, b\}$ already belongs to T_j and has degree at most three, so $p_y \leq 2$, contrary to $p_y \geq 3$.

Case 3: the outside parts are the same pair $\{x, x'\}$. The two members of $\mathcal{B}_y$ then use different anchor points. Pair degree leaves at most one petal from $\mathcal{F}_i$ equal to $\{x, x'\}$. The remaining petals contain x or x' . Call one an x -petal if it contains x but not x' , and define an x' -petal symmetrically. If there are at most three of each type, there are at most seven petals in total. Thus one type occurs at least four times; suppose that there are at least four x -petals.

If an x' -petal $\{x', v\}$ occurs in R_s , cross-intersection forces every x -petal in either of the other two residue graphs to equal $\{x, v\}$, so there is at most one in each. Since $\Delta(R_s) \leq 3$ and there are at least four x -petals altogether, R_s contains at least two distinct x -petals. These two edges prevent an x' -petal from occurring in either other residue graph. Moreover, at least one x -petal occurs outside R_s ; it is $\{x, v\}$ and forces every x' -petal in R_s to equal $\{x', v\}$. Hence there are at most five x -petals, one x' -petal, and one petal $\{x, x'\}$, again at most seven in total. Therefore no x' -petal occurs, and all eight petals contain x . The symmetric argument applies if the x' -petals are the type occurring at least four times.

Thus all eight one-point members of $\mathcal{F}_i$ and both members of $\mathcal{B}_y$ contain one common point, say x . Each of the six members counted by $p_y + r_y$ must contain x or x' in order to meet both members of $\mathcal{B}_y$. At most three contain x' , by the pair-degree bound for $\{y, x'\}$. Hence at least one additional member contains x , giving degree at least 11.

Case 4: the outside parts share exactly one point, say $\{w, c\}$ and $\{w, d\}$. Every petal from $\mathcal{F}_i$ contains w or equals $\{c, d\}$. At most two of the p_y two-point members of $\mathcal{F}_j$ and at most one of its r_y one-point members avoid w . Thus at least three of the six members counted by $p_y + r_y$ contain w .

If none of R_1, R_2, R_3 contains $\{c, d\}$, all eight petals from $\mathcal{F}_i$ contain w . Together with the two members of $\mathcal{B}_y$ and at least three further members of $\mathcal{F}_j$, this gives $d_{\mathcal{F}}(w) \geq 13$.

If at least two residue graphs contain $\{c, d\}$, cross-intersection forces every petal to be one of $\{c, d\}$, $\{w, c\}$, and $\{w, d\}$. The first pair can occur in at most three members of $\mathcal{F}_i$. Each of the other two pairs already occurs in one member of $\mathcal{B}_y$, so each can occur in at most two members of $\mathcal{F}_i$. Thus $\mathcal{F}_i$ has at most $3 + 2 + 2 = 7$ one-point members, contrary to $r = 8$.

Finally, suppose that exactly one residue graph contains $\{c, d\}$. Each of the other two graphs can then contain only $\{w, c\}$ and $\{w, d\}$, so has at most two edges. Reaching eight petals forces both graphs to contain both edges, while the graph containing $\{c, d\}$ contains three further edges through w . Hence seven members of $\mathcal{F}_i$ contain w . Together with the two members of $\mathcal{B}_y$ and at least three further members of $\mathcal{F}_j$, this gives $d_{\mathcal{F}}(w) \geq 12$. This excludes a non-rigid size-9 family $\mathcal{F}_j$.

Suppose finally that $\mathcal{F}_j$ is rigid. If $a_j = 4$, the assertion holds. Otherwise (ii), applied to $\mathcal{F}_j$, gives $p(\mathcal{F}_j, T_j) = 0$. The graph $L'_y(T_j)$ is the seven-edge graph R from lemma 5.5. Indeed, R is $K_{2,3}$ together with one edge inside its three-vertex part. Every triangle contains that additional edge, and the two vertices outside the triangle form an edge. Thus every triangle in R leaves exactly one disjoint edge. At most one one-point petal of $\mathcal{F}_j$ avoids the rigid triangle, and $d_{\mathcal{F}_j}(y) \leq 2$. $\square$

The profile variables satisfy the aggregate bound

$$\sum_{i=1}^3 V_i \leq 96. \quad (5.10)$$

If no V_i equals 33, this follows from integrality and equation (5.9). If $V_i = 33$, then $(N_i, a_i) = (25, 10)$. This rigid 10-block has $r = 9$ and common point in another anchor T_j . The cover table gives $|\mathcal{B}_y| \leq 1$. Since the members of $\mathcal{F}_j$ split according to whether they contain y ,

$$a_j = d_{\mathcal{F}_j}(y) + |\mathcal{B}_y|.$$

If $\mathcal{F}_j$ is rigid, lemma 5.7 gives either $a_j = 4$ or $d_{\mathcal{F}_j}(y) \leq 2$. Thus $a_j \leq 4$ and $V_j \leq 27$. If $\mathcal{F}_j$ is non-rigid, the same lemma gives $d_{\mathcal{F}_j}(y) \leq 7$, so $a_j \leq 8$; moreover, $N_j \leq 24$, and hence $V_j \leq 30$. Together with $V_k \leq 33$, both cases imply equation (5.10).

We now encode the local structure around each anchor by finitely many integer variables for use in the profile enumeration. Write an anchor as $T_j = \{v_0, v_1, v_2\}$. Let

- n_{uv} be the number of members meeting T_j exactly in $\{u, v\}$, and let q_{uv} count those members lying in $\mathcal{F}_j$;
- $L_v = |E(L'_v(T_j))|$, and let r_v count the corresponding one-point members lying in $\mathcal{F}_j$;
- k_v be the number of rigid blocks whose common point is v .

The profile verifier exhausts

$$0 \leq q_{uv} \leq n_{uv} \leq 2, \quad 0 \leq r_v \leq L_v \leq 7, \quad (5.11)$$

$$1 + \sum_{\{u,v\} \in \binom{T_j}{2}} n_{uv} + \sum_{v \in T_j} L_v = N_j, \quad 1 + \sum_{\{u,v\} \in \binom{T_j}{2}} q_{uv} + \sum_{v \in T_j} r_v = a_j, \quad (5.12)$$

$$1 + \sum_{u \in T_j \setminus \{v\}} n_{uv} + L_v \leq 10, \quad L_v \geq r_v + 3k_v. \quad (5.13)$$

If $k_v \geq 2$, the two induced triangles are vertex-disjoint and exhaust the link: any additional edge, together with one edge from each triangle chosen to avoid its endpoints, would form a three-matching. Hence $L_v = 6$ and $r_v = 0$. If $k_v \geq 1$, the petals counted by r_v avoid the induced triangle. Two disjoint such petals, together with an edge of the triangle, would again form a three-matching. They are therefore intersecting, and the degree bound gives $r_v \leq 3$. When $L_v = 7$, the link is the graph R from lemma 5.5, in which a triangle has only

one disjoint edge; thus $r_v \leq 1$. The number of members of $\mathcal{F}_j$ avoiding v is bounded by the applicable entry of lemma 5.7. For a rigid family $\mathcal{F}_j$, lemma 5.5 forces $n_{uv} = 1$, $L_v = 7$, and $q_{uv} = 0$ unless $a_j = 4$.

For each $v \in T_j$, form the pair

$$\left(\sum_{u \in T_j \setminus \{v\}} q_{uv}, r_v \right).$$

The first coordinate counts the non-anchor members of $\mathcal{F}_j$ that meet T_j in two points and contain v ; the second counts those meeting T_j exactly at v . We call the resulting triple of pairs the anchor signature. It must be the signature of an actual rooted block of size a_j . The complete block enumeration gives

$$2, 6, 13, 23, 35, 49, 51, 41, 16$$

signatures at sizes 2 through 10, respectively. All ranges used by the profile calculation are therefore finite.

Proposition 5.8 (Profile reduction). *Up to a simultaneous permutation of the three anchors, a hypothetical 50-member family has one of the following four profiles.*

	(a_1, a_2, a_3)	(c_1, c_2, c_3)	z	(N_1, N_2, N_3)
<i>I</i>	(8, 9, 10)	(6, 8, 9)	0	(25, 24, 24)
<i>II</i>	(8, 9, 10)	(7, 7, 9)	0	(24, 25, 24)
<i>III</i>	(9, 9, 10)	(7, 7, 8)	0	(24, 24, 24)
<i>IV</i>	(8, 9, 10)	(7, 8, 8)	0	(24, 24, 25)

Their V -vectors are, respectively,

$$(31, 31, 32), \quad (30, 32, 32), \quad (31, 31, 32), \quad (30, 31, 33).$$

Computer-assisted proof. Order the anchors so that $a_1 \leq a_2 \leq a_3$. Enumerate the integers $1 \leq a_i \leq 10$ and nonnegative c_1, c_2, c_3, z . Test the partition and derived identities in equations (5.5) and (5.7). Apply equation (5.9), the neighborhood and aggregate bounds, and the fact proved in lemma 5.9 below that the cross class between two rooted blocks of size 10 has at most two members. This leaves 55 numerical profiles. The conditions in lemma 5.7 leave 11. Next impose the local ranges in equation (5.11) and the identities in equation (5.12). The degree conditions in equation (5.13) and the complete anchor signatures leave exactly the four displayed profiles.

A separate verifier implements the same constraints, reproduces the counts 55, 11, and 4, and checks the four displayed profiles against the cross-family bounds below. $\square$

5.5 Cross-family bounds

To eliminate the four remaining profiles, we bound the number of members meeting two anchors in terms of the sizes and structures of the rooted blocks attached to them. This subsection is independent of the standing 50-member assumption: its bounds concern an arbitrary 3-uniform family with matching number two and no 4-sunflower. Theorem 5.10 applies them to the cross classes $\mathcal{C}_i$.

Let U_1, U_2 be a maximum two-matching in a 3-uniform family $\mathcal{H}$ without a 4-sunflower, where $\nu(\mathcal{H}) = 2$. Define

$$\begin{aligned}\mathcal{A}_1 &= \{H \in \mathcal{H} : H \cap U_1 \neq \emptyset, H \cap U_2 = \emptyset\}, \\ \mathcal{A}_2 &= \{H \in \mathcal{H} : H \cap U_2 \neq \emptyset, H \cap U_1 = \emptyset\}, \\ \mathcal{X} &= \{H \in \mathcal{H} : H \cap U_1 \neq \emptyset, H \cap U_2 \neq \emptyset\}.\end{aligned}$$

The families $\mathcal{A}_1, \mathcal{A}_2$ are rooted blocks. Let $\text{mc}(a, b)$ be the maximum of $|\mathcal{X}|$ under the conditions $|\mathcal{A}_1| \geq a$ and $|\mathcal{A}_2| \geq b$. Let $\text{mc}_0(a, b)$ be the same maximum with $p(\mathcal{A}_1, U_1) = 0$.

Deleting a non-anchor block member preserves the two anchors, all cross members, the absence of a 4-sunflower, and matching number 2. These quantities are therefore non-increasing in each block-size argument. Since rooted blocks have size at most 10, the exact lists at sizes 8, 9, 10 suffice for the values below.

Lemma 5.9 (Exact cross-family bounds). *One has*

$$\text{mc}_0(8, 10) \leq 7, \quad \text{mc}_0(9, 10) \leq 6, \quad \text{mc}(10, 9) \leq 6, \quad \text{mc}(10, 10) \leq 2. \quad (5.14)$$

Computer-assisted proof. The 10-by-10 calculation checks all 6555 unordered pairs of the 114 rooted 10-block classes. No pair admits a cross class of size 3.

For the other bounds, use the complete rooted lists in lemma 5.4. Make the two anchors disjoint. Every outside point of the second block is either identified injectively with an unused outside point of the first block or assigned the next fresh label. This enumerates every possible intersection of the outside supports, since such an identification is a partial bijection.

A partial identification is rejected only if a necessary condition fails: distinctness, point degree at most 10, pair degree at most 3, or an already completed 4-sunflower. After a full identification, the program generates every possible cross triple. It contains a point of each anchor; its third point is in the merged block support or is fresh. A target of t cross members needs at most t fresh labels, since each cross triple has at most one point outside the merged support. Fresh labels are interchangeable.

The subset recursion tests distinctness, every three-matching, and every four-subfamily for the sunflower condition. Its symmetry reduction orders the images of interchangeable degree-one outside twins with the same other pair. These twins are freely permutable, so one representative of each orbit remains. The recursion retains merged base pairs that already contain a three-matching; the resulting search space therefore contains every valid pair.

The exhaustive counts are

bound	rooted pairs	cross-class size excluded
$\text{mc}_0(8, 10) \leq 7$	$45 \cdot 114 = 5130$	8
$\text{mc}_0(9, 10) \leq 6$	$27 \cdot 114 = 3078$	7
$\text{mc}(10, 9) \leq 6$	$665 \cdot 114 = 75810$	7

The first two rows prove the two restricted inequalities. The last row handles exact sizes 9 and 10. The only larger case in $\text{mc}(10, 9)$ is 10 by 10, already bounded by 2. $\square$

5.6 Exclusion of the profiles

Theorem 5.10. *Every 3-uniform family with 50 distinct members contains a 4-sunflower. Consequently,*

$$f(3, 4) \leq 49.$$

Proof. Assume that $\mathcal{F}$ has 50 members and contains no 4-sunflower. Use the maximum-matching decomposition above. By proposition 5.8, after permuting the anchors there are four cases.

For $\{i, j, k\} = \{1, 2, 3\}$, the subfamily $\mathcal{F}_j \cup \mathcal{F}_k \cup \mathcal{C}_i$ has matching number exactly 2. It contains T_j, T_k , while three disjoint members together with T_i would form a four-matching. Thus $\mathcal{C}_i$ is a cross class to which lemma 5.9 applies.

In profile I, $N_1 = 25$, so $\mathcal{F}_1$ is rigid. It has size 8, and lemma 5.7(ii) gives $p(\mathcal{F}_1, T_1) = 0$. The class $\mathcal{C}_2$ is the cross class between $\mathcal{F}_1$ and $\mathcal{F}_3$, of sizes 8 and 10. Therefore

$$c_2 \leq \text{mc}_0(8, 10) \leq 7,$$

contrary to $c_2 = 8$.

In profile II, $N_2 = 25$, so the size-9 block $\mathcal{F}_2$ is rigid and $p(\mathcal{F}_2, T_2) = 0$. The class $\mathcal{C}_1$ is the cross class between $\mathcal{F}_2$ and the size-10 block $\mathcal{F}_3$. Hence

$$c_1 \leq \text{mc}_0(9, 10) \leq 6,$$

contrary to $c_1 = 7$.

In profiles III and IV, $\mathcal{C}_1$ is the cross class between blocks of sizes 9 and 10. In each case,

$$c_1 \leq \text{mc}(10, 9) \leq 6,$$

contrary to $c_1 = 7$. All four profiles are impossible. $\square$

Combining theorems 5.1 and 5.10 gives

$$39 \leq f(3, 4) \leq 49.$$

5.7 Computational verification

Four computations enter theorem 5.10: the rooted-block enumeration in lemma 5.4, the eight-vertex cover enumeration in lemma 5.7, the profile calculation in proposition 5.8, and the cross-family searches in lemma 5.9. The supplementary material contains the enumerators, the canonicalizer, the profile verifier, the search results, and the result checker. The checker reproduces

$$55 \longrightarrow 11 \longrightarrow 4$$

profiles, verifies the cover bounds 3, 2, 1 for $r = 7, 8, 9$, and checks all 5130, 3078, and 75810 rooted-pair results. The enumerators establish the exclusions; the checker separately verifies that their result sets cover all required rooted pairs.

6 Bounds for $f(4, 3)$

This section proves $54 \leq f(4, 3) \leq 83$ and determines the intersecting extremum $\psi(4, 3, 2) = 27$ exactly. For the upper bound, we fix a member R and split the family into the members meeting R and the members disjoint from R . The members disjoint from R pairwise intersect, so their number is at most $\psi(4, 3, 2)$. The members meeting R number at most 56, by an analysis of the exact trace classes of R . Together these bounds give $f(4, 3) \leq 56 + 27 = 83$. The lower bound places two copies of the extremal 27-member family on disjoint supports. The upper bound is computer-assisted at two points. The support bound in proposition 6.1 rests on fifteen unsatisfiability certificates, and the inequality $\psi(4, 3, 2) \leq 27$ rests on an exhaustive canonical-augmentation search.

6.1 Finite triple inputs

The neighborhood argument uses two finite facts about intersecting triple families. The first is a size bound. The second concerns the support at equality.

The following exact value is due to Abbott and Gardner [2]:

$$f(3, 3) = 20. \quad (6.1)$$

Proposition 6.1 (Finite triple bounds). *The following statements hold.*

- (i) *Every intersecting, 3-uniform family without a 3-sunflower has at most 10 members. In the notation of equation (2.3),*

$$\psi(3, 3, 2) \leq 10.$$

- (ii) *If such a family has exactly 10 members, then its support has at most 6 points.*

Proof. Part (i), due to Abbott and Hanson [4, p. 9], follows directly from equations (3.5) and (6.1), which give

$$20 = f(3, 3) \geq 2\psi(3, 3, 2).$$

We prove part (ii) by a finite computation. Fix the anchor $\{0, 1, 2\}$ and use one Boolean variable for each triple that meets it. A sequential counter imposes the cardinality of exactly ten selected triples [17], and the anchor variable is selected. For each pair of disjoint triples, a two-literal clause forbids selecting both. For every pair of points, an at-most-two constraint excludes a sunflower with that pair as core. For every point, a clause excludes each three-matching in its link. This excludes a sunflower with a one-point core. Pairwise intersection already excludes the empty-core case. These are all possible core sizes for three distinct triples. For an exact-support instance, each point is also required to occur in a selected triple.

An intersecting family of ten triples has support at most $3 + 9 \cdot 2 = 21$. The fifteen exact-support instances for support sizes $7, 8, \dots, 21$, each with exactly ten selected triples, are unsatisfiable. The fifteen unsatisfiability proofs were generated by CaDiCaL [8] and accepted by the independent checker `drat-trim` [18]. The CNF encoder generates the clauses described above directly. The anchor and support bounds show that the instances cover every family in part (ii). $\square$

Equation (2.6) gives $f(2, 3) = 6$. An intersecting graph without a 3-sunflower has at most three edges: an intersecting simple graph is a star or a triangle, and a three-edge star is a sunflower. Hence

$$\psi(2, 3, 2) = 3, \quad \psi(1, 3, 2) = 1. \quad (6.2)$$

The first value is also recorded by Abbott and Hanson [4, p. 9].

6.2 The closed meeting neighborhood

Let $\mathcal{F}$ be a 4-uniform family without a 3-sunflower, and fix $R \in \mathcal{F}$. Define

$$\mathcal{N}_{\mathcal{F}}[R] = \{F \in \mathcal{F} : F \cap R \neq \emptyset\}.$$

For $\emptyset \neq T \subsetneq R$, let

$$\mathcal{C}_T(R) = \{F \in \mathcal{F} : F \cap R = T\}$$

be the exact trace class at T .

Lemma 6.2 (Exact-trace residues). *The residues $F \setminus T$, $F \in \mathcal{C}_T(R)$, are pairwise distinct and form an intersecting, $(4 - |T|)$ -uniform family without a 3-sunflower. Consequently,*

$$|\mathcal{C}_T(R)| \leq \psi(4 - |T|, 3, 2). \quad (6.3)$$

Proof. Exactness gives $(F \setminus T) \cap R = \emptyset$. Since $T \subseteq F$ for every $F \in \mathcal{C}_T(R)$, adjoining T recovers F ; distinct members therefore give distinct residues. Two disjoint residues, together with R , would lift to a sunflower with core T . A sunflower among three residues would also lift after adjoining T . $\square$

For $x \in R$ and $\{x, y\} \in \binom{R}{2}$, put

$$\begin{aligned} s_x &= |\mathcal{C}_{\{x\}}(R)|, & s &= \sum_{x \in R} s_x, \\ p_{xy} &= |\mathcal{C}_{\{x, y\}}(R)|, & p &= \sum_{\{x, y\} \in \binom{R}{2}} p_{xy}, & r_x &= \sum_{y \in R \setminus \{x\}} p_{xy}, \\ t_m &= |\mathcal{C}_{R \setminus \{m\}}(R)|, & t &= \sum_{m \in R} t_m. \end{aligned}$$

By proposition 6.1, lemma 6.2, and equation (6.2),

$$s_x \leq 10, \quad p_{xy} \leq 3, \quad t_m \leq 1. \quad (6.4)$$

Put $h = s + p + t$. The only member with full trace R is R itself, so

$$|\mathcal{N}_{\mathcal{F}}[R]| = 1 + h. \quad (6.5)$$

The link bound and equation (6.1) give $d_{\mathcal{F}}(x) \leq 20$ for every point x . Counting incidences with the four points of R , apart from the four incidences contributed by R , gives

$$s + 2p + 3t \leq 4(20 - 1) = 76. \quad (6.6)$$

Call $x \in R$ saturated if $s_x = 10$, and define

$$\mathcal{H}_x = \{F \setminus \{x\} : F \in \mathcal{C}_{\{x\}}(R)\}, \quad U_x = \text{supp}(\mathcal{H}_x).$$

If x is saturated, proposition 6.1 gives $|U_x| \leq 6$. No pair occurs in three members of $\mathcal{H}_x$. Counting the pairs in its ten triples gives

$$30 \leq 2 \binom{|U_x|}{2} \leq 30.$$

Equality holds in both bounds. Thus $|U_x| = 6$, every pair of support points has degree 2, and every support point has degree 5. In particular, $\mathcal{H}_x$ is a simple 2 -($6, 3, 2$) design. Five blocks contain any fixed support point, and two blocks contain any fixed pair. Thus two of the ten blocks avoid any prescribed pair of support points. In particular, a block avoids any prescribed set of at most two support points.

Lemma 6.3 (Incident pair traces). *If x is saturated, then $r_x \leq 6$.*

Proof. For $y \in R \setminus \{x\}$, the residues of $\mathcal{C}_{\{x, y\}}(R)$ are distinct pairwise-intersecting edges. If $p_{xy} = 3$, they cannot form a three-edge star, since their lifts would form a sunflower. They therefore form a triangle, with support V_y .

Consider the complete link at x . Every point in this triple family has degree at most $f(2, 3) = 6$. A point in $U_x \cap V_y$ would occur five times in $\mathcal{H}_x$ and twice in the y -triangle. Hence

$$U_x \cap V_y = \emptyset \quad \text{when } p_{xy} = 3. \quad (6.7)$$

Thus every edge of the y -triangle is disjoint from U_x .

Suppose that $r_x \geq 7$. Write $R \setminus \{x\} = \{y, z, w\}$ and choose the labels so that $p_{xy} = 3$. Then $p_{xz} + p_{xw} \geq 4$.

If $p_{xz} = 3$ or $p_{xw} = 3$, relabel so that $p_{xz} = 3$, and let Δ be the y -triangle. Every edge of the z -triangle meets every edge of Δ . Otherwise those two disjoint edges would allow us to choose a block of $\mathcal{H}_x$ avoiding the at most two points of the z -edge that lie in U_x . The y -edge is disjoint from U_x , so these would be three disjoint triples in the link at x . An edge meeting all three edges of a triangle is an edge of that triangle, so the two triangles coincide. The w -class is nonempty, since $r_x \geq 7$. If one of its residue edges missed an edge of Δ , choose a block of $\mathcal{H}_x$ avoiding its at most two points in U_x . This would again give three disjoint link triples. The residue edge therefore belongs to Δ . It then occurs under all three labels y, z, w , giving a sunflower in the link.

The remaining case is $(p_{xz}, p_{xw}) = (2, 2)$. The same disjointness argument places both two-edge residue families inside the three edges of Δ . Two two-element subsets of a three-element set share an edge. That edge occurs under y, z, w and gives the same contradiction. $\square$

Let σ be the number of saturated points of R . Then

$$s \leq 10\sigma + 9(4 - \sigma) = 36 + \sigma, \quad \sigma \geq s - 36.$$

For a saturated point, lemma 6.3 gives $r_x \leq 6$; for every other point, equation (6.4) gives $r_x \leq 9$. Since each pair trace is counted at both endpoints,

$$2p = \sum_{x \in R} r_x \leq 6\sigma + 9(4 - \sigma) \leq 144 - 3s. \quad (6.8)$$

Substitute $s = h - p - t$ into equations (6.6) and (6.8). This gives

$$p + 2t \leq 76 - h, \quad p + 3t \geq 3h - 144.$$

Hence $t \geq 4h - 220$. Since $t \leq 4$, one has $h \leq 56$. Equality forces

$$(s, p, t) = (40, 12, 4). \quad (6.9)$$

Excluding this profile improves the bound on $|\mathcal{N}_{\mathcal{F}}[R]| = 1 + h$ from 57 to 56, and hence the final upper bound on $f(4, 3)$ from 84 to 83.

Lemma 6.4 (Exclusion of the equality profile). *The profile in equation (6.9) cannot occur.*

Proof. Assume equation (6.9). Every singleton class is saturated, and every triple trace is occupied once. Write

$$T_m = (R \setminus \{m\}) \cup \{z_m\}, \quad z_m \notin R \quad (m \in R).$$

For $x \in R$, define

$$\mathcal{K}_x = \{F \setminus \{x\} : x \in F \in \mathcal{F}, |F \cap R| \geq 2\}.$$

Every member $K \in \mathcal{K}_x$ contains at least one point of $R \setminus \{x\}$. The support U_x is disjoint from R . Hence $|K \cap U_x| \leq 2$. If the intersection is a pair, the two design blocks through that pair, together with K , form a sunflower. If it is a singleton $\{u\}$, the five design blocks through u induce a 2-regular simple graph on the other five support points. This graph is a 5-cycle. Two disjoint edges of the cycle give two design blocks meeting exactly in u ; together with K they form a sunflower. Thus

$$K \cap U_x = \emptyset \quad (K \in \mathcal{K}_x). \quad (6.10)$$

The family $\mathcal{K}_x$ is intersecting. Two disjoint members, together with any block of $\mathcal{H}_x$, would be three disjoint triples in the link at x . Counting over all $x \in R$ gives

$$\sum_{x \in R} |\mathcal{K}_x| = 4 + 2p + 3t = 40.$$

Each $\mathcal{K}_x$ is an intersecting triple family without a 3-sunflower, so proposition 6.1 bounds it by 10. Hence every $\mathcal{K}_x$ has size 10. By proposition 6.1(ii), its support has at most six points. No pair occurs in three members of $\mathcal{K}_x$, since those members would form a sunflower with that pair as core. The ten triples contain 30 pair occurrences. Therefore

$$30 \leq 2 \binom{|\text{supp}(\mathcal{K}_x)|}{2} \leq 30.$$

Equality holds in both bounds. Thus the support has exactly six points, every pair of support points has degree 2, and $\mathcal{K}_x$ is a simple 2-(6, 3, 2) design.

Fix x and put $Y = R \setminus \{x\}$. The design $\mathcal{K}_x$ contains

$$Y, \quad B_m = (Y \setminus \{m\}) \cup \{z_m\} \quad (m \in Y).$$

Its support has six points, so the three z_m , $m \in Y$, lie in the three-point complement of Y . They are distinct. Indeed, suppose that $z_i = z_j = q$ and let y be the third point of Y . The pairs $\{y, i\}$, $\{y, j\}$, and $\{y, q\}$ already occur twice among the displayed blocks. The two remaining design blocks through y must avoid i, j, q . Both would therefore consist of y and the other two outside support points, contradicting distinctness.

Write $Y = \{y, a, b\}$. The two design blocks through y not displayed above come from $\mathcal{C}_{\{x, y\}}(R)$. The pair $\{y, z_y\}$ occurs in none of the displayed blocks, so both new blocks contain z_y . The pairs $\{y, z_a\}$ and $\{y, z_b\}$ each occur once and each needs one further occurrence. Their residue edges are therefore

$$\{\{z_y, z_a\}, \{z_y, z_b\}\}. \quad (6.11)$$

For any two indices in R , choose x outside them. The preceding argument shows that their z -points are distinct. Thus all four points z_m are distinct.

Finally fix distinct x, y and write $R \setminus \{x, y\} = \{a, b\}$. The x -view of the exact pair class is equation (6.11). Its y -view is

$$\{\{z_x, z_a\}, \{z_x, z_b\}\}.$$

Both views describe

$$\{F \setminus \{x, y\} : F \in \mathcal{C}_{\{x, y\}}(R)\},$$

so they must be equal. They cannot be equal because the four z -points are distinct. $\square$

Proposition 6.5 (Meeting-neighborhood bound). *For every $R \in \mathcal{F}$,*

$$|\mathcal{N}_{\mathcal{F}}[R]| \leq 56. \quad (6.12)$$

Proof. The preceding inequalities give $h \leq 56$, and lemma 6.4 excludes equality. Thus $h \leq 55$. Apply equation (6.5). $\square$

6.3 The intersecting extremum

To determine $\psi(4, 3, 2)$, it remains to prove its upper bound. This follows from an exhaustive enumeration of the objects defined below.

The decision problem asks whether there is a family with exactly 28 distinct members, called rows below, satisfying the following conditions:

1. every row has size four;
2. every two rows intersect;
3. no three rows have equal pairwise intersections.

The ground set is not fixed. If such a family existed, its support would have size at most

$$4 + 27 \cdot 3 = 85, \tag{6.13}$$

because every row after the first meets the first row.

A partial family is represented by its two-colored bipartite incidence graph. Row vertices and point vertices are the two color classes. The first row is fixed as $\{0, 1, 2, 3\}$. If a parent uses the point labels $0, \dots, C-1$, a candidate row is generated by choosing an s -subset of the old labels, for $0 \leq s \leq 4$, and adjoining the fresh consecutive labels

$$C, C+1, \dots, C+(4-s)-1.$$

This list contains every extension up to relabelling points not used by the parent.

A candidate is rejected if it repeats a row, misses a prior row, or completes a 3-sunflower. For prior rows A, B and candidate S , the last test is

$$A \cap B \subseteq S, \quad S \cap A \subseteq B, \quad S \cap B \subseteq A.$$

These three inclusions are equivalent to $A \cap B = A \cap S = B \cap S$. All rejection conditions are hereditary under row deletion.

The program uses nauty to compute canonical labels, vertex orbits, and automorphisms of the colored incidence graph [15, 16]. Parent automorphisms reduce the candidate rows to orbit representatives. After a candidate is appended, the program selects the last row in the canonical row order. The child is accepted when the appended row belongs to the same full child-automorphism orbit as this canonical row. Deleting a row in that orbit, and then deleting points used only by that row, defines the canonical parent.

Lemma 6.6 (Canonical construction coverage). *Every isomorphism class of valid partial families occurs on an accepted construction path. Every valid 28-member family, if one exists, lies below an emitted valid depth-6 root.*

Proof. Proceed by induction on the number of rows. The one-row family has the fixed representative $\{0, 1, 2, 3\}$. Let $\mathcal{G}$ be a valid child. Delete a row in the orbit selected by its canonical labeling, together with points used only by that row. The resulting canonical parent $\mathcal{P}$ remains valid.

By induction, a normalized copy of $\mathcal{P}$ occurs. The deleted row is an old-label subset together with fresh points, so an isomorphic candidate is generated. Parent-orbit reduction retains a representative of its orbit, and the appended row lies in the canonical deletion orbit of the resulting child. The child is therefore accepted.

The implementation stores at most 400 generator permutations. The stored automorphisms generate a subgroup of the full parent automorphism group, and subgroup orbits refine full orbits. The limit may duplicate work, but it preserves at least one representative of every extension. The canonical-deletion test uses nauty's full orbit output rather than the stored generator list.

Every accepted object deeper than six rows has an accepted depth-6 ancestor obtained by repeated canonical deletion. This proves the second statement. $\square$

The implementation permits 128 point columns, 56 rows, and 184 vertices in the incidence graph. A target needs at most $28 + 85 = 113$ vertices, so these capacities do not restrict the decision problem.

The depth-6 generation visits 67,193 search-tree nodes and produces 11,720 roots. Exhausting the subtrees below these roots visits 105,917,089,577 search-tree nodes and finds no 28-member family. A separate C verifier reconstructs the depth-6 frontier and checks that the results contain one completed record for every root, with the stated total node count.

Proposition 6.7 (Intersecting upper bound). *Every intersecting, 4-uniform family without a 3-sunflower has at most 27 members.*

Computer-assisted proof. Suppose that a 28-member family existed. Equation (6.13) places it within the program capacities. By lemma 6.6, an isomorphic copy lies below one of the 11,720 depth-6 roots. The exhaustive search of every such subtree finds no target. This is a contradiction. A larger family would contain a 28-member subfamily. $\square$

For the reverse inequality, partition nine points into three disjoint three-point sets X_0, X_1, X_2 . For every unordered pair $\{i, j\} \subseteq \{0, 1, 2\}$, take all sets

$$A_i \cup A_j, \quad A_i \in \binom{X_i}{2}, \quad A_j \in \binom{X_j}{2}. \quad (6.14)$$

There are $3 \cdot 3 \cdot 3 = 27$ such sets. This is the construction of Abbott and Hanson [4, p. 9].

Proposition 6.8 (Intersecting extremum). *One has*

$$\psi(4, 3, 2) = 27. \quad (6.15)$$

Proof. Two members of the family in equation (6.14) use two of the three point sets, so their types share one point set. Any two pairs in a three-point set meet. The family is therefore intersecting.

Consider three members. If their three point-set types are equal, equal pairwise intersections in each used point set would force all three chosen pairs to be equal. The three members would then be equal. If exactly two types are equal, one of the two point sets used by those members is not used by the third. The two members intersect in that point set, whereas neither of the other two intersections contains a point from it. If the types are X_0X_1 , X_0X_2 , and X_1X_2 , their three pairwise intersections are nonempty subsets of X_0, X_1, X_2 , respectively. They are not equal. Thus the family contains no 3-sunflower.

This proves the lower bound 27. The reverse inequality is proposition 6.7. $\square$

6.4 The global bounds

Take two copies of the family in equation (6.14) on disjoint supports. Their union has 54 members. Three members chosen from one copy do not form a sunflower. If three members use both copies, two lie in the same copy and meet, whereas their intersections with a member of the other copy are empty. Thus their three pairwise intersections are not equal.

Theorem 6.9. *One has*

$$54 \leq f(4, 3) \leq 83. \quad (6.16)$$

Proof. The preceding two-copy construction gives the lower bound. For the upper bound, let $\mathcal{F}$ be a 4-uniform family without a 3-sunflower and choose $R \in \mathcal{F}$. Put

$$D_{\mathcal{F}}(R) = \{F \in \mathcal{F} : F \cap R = \emptyset\}.$$

If two members of $D_{\mathcal{F}}(R)$ were disjoint, those two members and R would form a sunflower with empty core. Hence $D_{\mathcal{F}}(R)$ is intersecting. By propositions 6.5 and 6.8,

$$|\mathcal{F}| = |\mathcal{N}_{\mathcal{F}}[R]| + |D_{\mathcal{F}}(R)| \leq 56 + 27 = 83.$$

□

6.5 Computational verification

Two finite computations enter the upper bound in theorem 6.9. First, part (ii) of proposition 6.1 uses fifteen CNF instances whose unsatisfiability proofs are checked by `drat-trim`; the anchor and support bounds in its proof establish coverage. Second, proposition 6.7 uses canonical augmentation, with coverage proved in lemma 6.6, to exhaust all 11,720 depth-6 roots. The supplementary material contains the search source, the manifests, the results, the frontier verifier, and the control programs. The search establishes the exclusion below each root. The verifier reconstructs the frontier and checks one completed result for each root and the total node count.

7 Lower bounds for six and seven petals

This section proves $f(3, 6) \geq 153$ and $f(3, 7) \geq 259$. Combined with corollary 4.3, these bounds give $153 \leq f(3, 6) \leq 255$ and $259 \leq f(3, 7) \leq 474$. Both constructions place two families on disjoint supports: a first component with matching number at most $k - 2$, and an intersecting second component. For six petals the first component is a cyclic family of 126 triples on 14 points; for seven petals it is a family of 217 triples on 17 points. The second components have 27 and 42 members, respectively.

Equation (3.6) gives

$$\psi(3, 6, 2) = 27, \quad \psi(3, 7, 2) = 42.$$

Thus the 27- and 42-member intersecting families used as the second components below are optimal. Within these two-component disjoint-support constructions, any further improvement must come from the first component.

7.1 Six petals

Work in $\mathbb{Z}_{14}$. Let $\mathcal{H}$ consist of all translates modulo 14 of the nine triples

$$\begin{aligned} &\{0, 1, 5\}, \{0, 1, 6\}, \{0, 1, 7\}, \{0, 1, 9\}, \{0, 1, 10\}, \\ &\{0, 10, 12\}, \{0, 2, 6\}, \{0, 2, 9\}, \{0, 2, 10\}. \end{aligned} \tag{7.1}$$

Lemma 7.1. *The family $\mathcal{H}$ has 126 members. It is 27-regular, has matching number at most 4, and contains no 6-sunflower.*

Proof. The stabilizer of a three-element subset of $\mathbb{Z}_{14}$ has order dividing both 3 and 14. It is therefore trivial, so every seed has an orbit of length 14. The cyclic gap sequences of the nine seeds, up to cyclic rotation, are

$$\begin{aligned} &(1, 4, 9), (1, 5, 8), (1, 6, 7), (1, 8, 5), (1, 9, 4), \\ &(10, 2, 2), (2, 4, 8), (2, 7, 5), (2, 8, 4). \end{aligned}$$

They are distinct. Hence the nine orbits are disjoint and $|\mathcal{H}| = 9 \cdot 14 = 126$.

Five disjoint triples would require 15 points. Thus $\nu(\mathcal{H}) \leq 4$. To exclude two-point cores, count the pairs in the nine seeds according to cyclic distance. Translation gives the following pair degrees. A pair at cyclic distance 7 is fixed by the translation by 7, so its orbit has length 7, and each seed pair at that distance contributes twice to the degree.

cyclic distance	1	2	3	4	5	6	7
pair degree in $\mathcal{H}$	5	5	0	5	5	5	4

No pair is therefore the core of a 6-sunflower.

For a one-point core, consider $\text{Lk}_{\mathcal{H}}(0)$. The vertices 3 and 11 are isolated because pairs at cyclic distance 3 have degree zero. Every edge of the link lies on the other 11 vertices, so its matching number is at most 5. Translation gives the same conclusion for every point. This excludes all possible core sizes.

Translation also makes all point degrees equal. Incidence counting gives

$$14d_{\mathcal{H}}(0) = 3|\mathcal{H}| = 378,$$

so every point has degree 27. □

The construction attains the maximum possible size on a 14-point support.

Proposition 7.2. *Every 3-uniform family on at most 14 points that contains no 6-sunflower has at most 126 members.*

Proof. Let $\mathcal{A}$ be such a family. For every point x , the link graph $\text{Lk}_{\mathcal{A}}(x)$ has maximum degree at most 5 and matching number at most 5. A violation gives a 6-sunflower with core of size two or one, respectively. By the Chvátal–Hanson formula in (2.5),

$$|E(\text{Lk}_{\mathcal{A}}(x))| \leq \text{CH}(5, 5) = 27.$$

Summing the point degrees gives

$$3|\mathcal{A}| \leq 14 \cdot 27 = 378.$$

Thus $|\mathcal{A}| \leq 126$. □

For a second component, take the graph G on a disjoint copy of $\mathbb{Z}_{11}$ with edge set

$$\{\{i, i+1\}, \{i, i+2\} : i \in \mathbb{Z}_{11}\} \cup \{\{i, i+5\} : 0 \leq i \leq 4\}, \quad (7.2)$$

where addition in the first set is modulo 11. The graph has 27 edges, maximum degree 5, and matching number at most 5. It therefore contains no 6-sunflower. Add a new point c and put

$$\mathcal{K} = \{\{c\} \cup e : e \in E(G)\}.$$

The family $\mathcal{K}$ contains no 6-sunflower, has 27 members, and has matching number 1.

Theorem 7.3. *One has*

$$153 \leq f(3, 6) \leq 255.$$

Proof. Place $\mathcal{H}$ and $\mathcal{K}$ on disjoint supports. A sunflower with nonempty core cannot use members from both families. An empty-core sunflower is a matching, while

$$\nu(\mathcal{H} \dot{\cup} \mathcal{K}) = \nu(\mathcal{H}) + \nu(\mathcal{K}) \leq 4 + 1 = 5.$$

Hence $\mathcal{H} \dot{\cup} \mathcal{K}$ contains no 6-sunflower and has $126 + 27 = 153$ members. The upper bound is corollary 4.3. $\square$

7.2 Seven petals

The next construction uses a finite triple system. The family and a standalone verifier are part of the supplementary material.

The verifier computes graph matching numbers by the following direct recurrence. For a graph G and $S \subseteq V(G)$, put $\mu_G(S) = \nu(G[S])$ and $\mu_G(\emptyset) = 0$. For nonempty S , choose $v \in S$. Then

$$\mu_G(S) = \max(\{\mu_G(S \setminus \{v\})\} \cup \{1 + \mu_G(S \setminus \{u, v\}) : u \in N_G(v) \cap S\}). \quad (7.3)$$

The two terms distinguish whether a maximum matching leaves v unmatched or matches it to a neighbor. Memoization over vertex subsets makes the calculation exact.

Lemma 7.4. *There is a family $\mathcal{A}$ of 217 triples on the point set $\{0, 1, \dots, 16\}$ such that every pair has degree at most 6 and every vertex link has matching number at most 6. The family contains no 7-sunflower and satisfies $\nu(\mathcal{A}) \leq 5$.*

Proof. Take the family listed in the supplementary material. The verifier checks that it contains 217 distinct sorted triples and has support $\{0, 1, \dots, 16\}$. It enumerates all pair degrees and applies (7.3) to every vertex link. The resulting link data are

vertex	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$ \text{Lk}_{\mathcal{A}}(v) $	38	39	39	39	38	39	39	38	35	38	37	39	38	39	38	39	39
$\nu(\text{Lk}_{\mathcal{A}}(v))$	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6

A sunflower of triples has core size zero, one, or two. A two-point core would require pair degree at least 7. A one-point core would require a matching of size 7 in a vertex link. Seven disjoint triples cannot fit on 17 points. These facts exclude the three cases. The point count also gives $\nu(\mathcal{A}) \leq \lfloor 17/3 \rfloor = 5$. $\square$

Take two disjoint seven-point sets P, Q and a new point z , all outside the support of $\mathcal{A}$. Let K_P and K_Q be the complete graphs on P and Q , respectively, and put

$$H = K_P \dot{\cup} K_Q, \quad \mathcal{B} = \{\{z, x, y\} : xy \in E(H)\}.$$

The graph H has maximum degree 6, matching number 6, and 42 edges. It contains no 7-sunflower. Deleting z shows that $\mathcal{B}$ also contains no 7-sunflower. Moreover, $|\mathcal{B}| = 42$ and $\nu(\mathcal{B}) = 1$.

Theorem 7.5. *One has*

$$259 \leq f(3, 7) \leq 474.$$

Proof. The supports of $\mathcal{A}$ and $\mathcal{B}$ are disjoint. A sunflower with nonempty core cannot use triples from both families. An empty-core sunflower is a matching, while

$$\nu(\mathcal{A} \dot{\cup} \mathcal{B}) = \nu(\mathcal{A}) + \nu(\mathcal{B}) \leq 5 + 1 = 6.$$

Thus $\mathcal{A} \dot{\cup} \mathcal{B}$ contains no 7-sunflower and has $217 + 42 = 259$ members. The upper bound is corollary 4.3. $\square$

7.3 Verification of the finite witness

The certificate for the lower bound $f(3, 7) \geq 259$ consists of the explicit 217-triple family and the 42-member family $\mathcal{B}$ constructed above. The standalone verifier checks the size and distinctness of the 217-triple family, its support, every pair degree, and every vertex-link matching number.

8 Formal verification in Lean

The main combinatorial arguments of sections 2 to 4, the appendix capacity recurrence, and every explicit finite witness used for the numerical lower bounds have been formalised in the Lean 4 proof assistant [10] over the Mathlib library [14]. Since the finiteness of $f(w, k)$ rests on the Erdős–Rado theorem, the formalisation represents f and ψ by pairs of upper-bound and witness predicates. The finite witness families are encoded explicitly in Lean, and their properties are verified by computation within Lean’s kernel.

Some external results and analytic arguments used in the paper are outside the Lean formalisation. The Chvátal–Hanson upper bound is supplied as a hypothesis. For the three exponential-rate bounds, Lean verifies constructions of size at least cr_k^w , for some $c > 0$ and every $w \geq 3$; the limiting and asymptotic arguments are not included. The exhaustive computations behind $f(3, 4) \leq 49$ and $f(4, 3) \leq 83$ are checked separately, as described in sections 5 and 6. The correspondence between the paper statements and formal theorems, together with the pinned toolchain and build instructions, is part of the supplementary material.

A Alternative capacity recurrence

This appendix proves a second lower-bound recurrence for $\psi(w, k, m)$, valid for every matching cap $2 \leq m \leq k - 1$. The construction assigns a smaller sunflower-free family to each edge of a bounded-degree auxiliary graph on m points; bounding the vertex degrees limits how often

any auxiliary point is used. At $m = k - 1$ the resulting bound coincides with corollary 3.2, so no result in the body of the paper depends on this appendix. At smaller matching caps neither recurrence dominates the other. For $k = 5$ and $m = 3$, theorem 3.1 gives $\psi(w, 5, 3) \geq 4\psi(w - 1, 5, 3) + f(w - 3, 5)$, while theorem A.2 gives $\psi(w, 5, 3) \geq 3\psi(w - 1, 5, 3) + 3f(w - 2, 5)$; which bound is larger depends on the relative growth of the two quantities.

We use the following standard circulant construction.

Lemma A.1 (Bounded-degree graph). *For integers $n \geq 1$ and $0 \leq d \leq n - 1$, there is a graph on n vertices with maximum degree at most d and exactly $\lfloor nd/2 \rfloor$ edges.*

Proof. Use vertex set $\mathbb{Z}_n$. If $d = 0$, take the empty graph. If d is positive and even, join vertices at cyclic distances $1, \dots, d/2$. The resulting graph is d -regular.

Suppose that d is odd and n is even. Join vertices at cyclic distances $1, \dots, (d - 1)/2$ and add the antipodal perfect matching. This graph is d -regular.

Suppose that both n and d are odd. Then $d \leq n - 2$. Start with the $(d - 1)$ -regular graph using cyclic distances $1, \dots, (d - 1)/2$. The edges of cyclic distance $(n - 1)/2$ form one cycle of length n , since

$$\gcd\left(\frac{n - 1}{2}, n\right) = 1.$$

Add a matching of size $(n - 1)/2$ from this cycle. The resulting graph has $n - 1$ vertices of degree d and one vertex of degree $d - 1$. It has $(nd - 1)/2 = \lfloor nd/2 \rfloor$ edges. $\square$

Theorem A.2 (Capacity recurrence). *Let $k \geq 3$ and $2 \leq m \leq k - 1$. Put*

$$d = \min\{k - m, m - 1\}, \quad E(m, k) = \left\lfloor \frac{md}{2} \right\rfloor.$$

For every $w \geq 3$,

$$\psi(w, k, m) \geq m\psi(w - 1, k, m) + E(m, k)f(w - 2, k). \quad (\text{A.1})$$

Proof. Take pairwise disjoint sets X, Y, Z , where $|Y| = m$. On X , take a $(w - 1)$ -uniform family $\mathcal{A}$ with no k -sunflower such that

$$|\mathcal{A}| = \psi(w - 1, k, m), \quad \nu(\mathcal{A}) \leq m - 1.$$

By lemma A.1, there is a graph P on Y with

$$\Delta(P) \leq d, \quad |E(P)| = E(m, k).$$

For each $p \in E(P)$, take a $(w - 2)$ -uniform family $\mathcal{E}_p$ on Z with no k -sunflower and with $|\mathcal{E}_p| = f(w - 2, k)$. Define

$$\begin{aligned} \mathcal{F} = & \{T \cup \{y\} : T \in \mathcal{A}, y \in Y\} \\ & \cup \{p \cup S : p \in E(P), S \in \mathcal{E}_p\}. \end{aligned} \quad (\text{A.2})$$

The two lines meet Y in one and two points. They are disjoint, and all members within each line are distinct. This gives the count in equation (A.1).

Let a matching in $\mathcal{F}$ contain q members from the first line and b from the second. If $b = 0$, the X -parts form a matching in $\mathcal{A}$, so $q \leq m - 1$. If $b \geq 1$, disjointness of their intersections with Y gives $q + 2b \leq m$. Hence

$$q + b \leq m - b \leq m - 1.$$

Thus $\nu(\mathcal{F}) \leq m - 1$.

Suppose that k members of $\mathcal{F}$ form a sunflower. Every member meets Y , and $|Y| = m < k$. By lemma 2.1, some $y \in Y$ belongs to all k members.

If all members come from the first line of equation (A.2), deleting y gives a k -sunflower in $\mathcal{A}$. Suppose that they all come from the second line. Every corresponding edge of P contains y . If a second endpoint occurs in two members, the occurrence rule forces it to occur in all k members. All members then use the same edge p , and deleting p gives a k -sunflower in $\mathcal{E}_p$. If no second endpoint repeats, the number of members is at most

$$\deg_P(y) \leq d \leq k - m < k.$$

Finally, suppose that both lines occur. The intersection of a member from the first line with one from the second is exactly $\{y\}$. This is the sunflower core. The X -parts of the members from the first line form a matching in $\mathcal{A}$, so there are at most $m - 1$ of them. A second endpoint of an edge of P cannot occur twice among the second-line members: it occurs in no member from the first line, contrary to the occurrence rule. There are therefore at most $\deg_P(y) \leq k - m$ members from the second line. The total is at most

$$(m - 1) + (k - m) = k - 1,$$

which is impossible. □

At $m = k - 1$, theorems A.2 and 3.1 both give equation (3.3). For other values of m , their coefficients and lags differ.

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